

Merged research report for the Surreal project

# Omnific Integers and Omnific–Diophantine Geometry

Retractions, rigidity, definability,  
and infinite families

Built from three manuscripts dated September 22, 2026

## Abstract

The omnific integers  $\mathbf{Oz} = \Pi \oplus \mathbb{Z}$  form a discretely ordered subring of the surreal field whose arithmetic is neither ordinary integer arithmetic nor a routine nonstandard enlargement of it. We organize their theory around the split ring retraction  $\text{ct} : \mathbf{Oz} \rightarrow \mathbb{Z}$  that takes a normal form to its constant coefficient. This gives an exact transfer theorem for polynomial equations with ordinary integer coefficients, a classification of finite quotients, and a complete solution criterion for linear systems.

The Diophantine results distinguish existence from the structure of solutions. At a nonzero constant level, the omnific fiber of a product of affine linear forms is described exactly by a real common kernel; in particular binary forms with at least two distinct projective linear factors, Pell equations and number-field norm forms have no new omnific points. Support-preserving Euler derivations prove the same rigidity for every separated equation  $ax^m + by^n = c$  with  $abc \neq 0$  and  $m, n \geq 2$ , including  $y^2 = x^3 + 1$ , and for nonzero Fermat triples of exponent at least three that generate the unit ideal. In contrast, every represented nonzero level of a nondegenerate indefinite integral quadratic form in at least three variables carries explicit injective polynomial families through each integer point, and we prove an exhaustive quadratic-level dichotomy that includes degenerate forms. Pell rigidity combined with the three-square or the four-square theorem gives single quartic equations defining the ordinary integers inside  $\mathbf{Oz}$ . Further results concern non-atomic factorization, full-class common divisors, bounded semialgebraic rigidity, primitive and projective points, a failed existential induction instance, and failures of residue lifting.

This report merges three independently written manuscripts; it states which source proved each result, prints shared results once, and keeps every source’s limitations. It is an AI-assisted draft: it has not been refereed, and nothing in it has been formalized in Lean.

Sources: manuscripts 01 (*Omnific Integers and Omnific–Diophantine Geometry*, the base text), 02 (*Omnific Integers and Diophantine Geometry*) and 05 (*Omnific Integers and Diophantine Rigidity*), all pinned to repository commit 2cb9c02. The provenance of every section is recorded in [appendix A](#); non-claims are collected in [appendix B](#). Finite checks accompany the report and check finite identities and examples only.

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# 1 Scope, results, and conventions

Omnific integers are the surreal numbers with no negative exponents in Conway normal form and with an ordinary integer constant coefficient. The definition is classical, as are their discreteness and integer-part property [1, 2, 3]. This report develops their arithmetic with a particular emphasis on polynomial equations.

Three questions must be kept separate. Does an equation have any solution? Does it have a genuinely infinite solution? Does it have such a solution with a nonvanishing, positivity or primitivity condition? For example, every system of equations over  $\mathbb{Z}$  has an omnific solution exactly when it has an ordinary integer solution. Nevertheless, the equation  $X^n + Y^n = Z^n$  has positive infinite omnific solutions for every ordinary  $n \geq 2$ . There is no contradiction: the constant coefficients of these solutions are all zero. On the other hand, every omnific solution of

$$x^2 - 2y^2 = 1, \quad y^2 = x^3 + 17, \quad x^7 - y^5 = 3$$

has ordinary integer coordinates; these conclusions concern the entire solution classes, not just their constant terms.

The central decomposition is

$$\mathbf{Oz} = \mathbb{Z} \oplus \Pi, \quad \text{ct} : \mathbf{Oz} \longrightarrow \mathbb{Z}, \quad \text{ct}(a + b) = \text{ct}(a) + \text{ct}(b), \quad \text{ct}(ab) = \text{ct}(a) \text{ct}(b), \quad (1.1)$$

where  $\Pi$  is the ideal of normal forms supported on strictly positive exponents; coefficients of elements of  $\Pi$  may have either sign. The decomposition is a splitting of additive groups and a ring retraction, not a direct product decomposition of rings. In particular,  $\Pi$  has no identity.

## 1.1 Principal results

The structural results of Sections 2–4 explain why ordinary congruences see only the constant coefficient, although the fraction field of the full class  $\mathbf{Oz}$  is all of  $\mathbf{No}$ . The main Diophantine statements are the following. Bracketed numbers name the source manuscripts (section 1.2).

- (1) **Equational transfer** [01, 02, 05]. For every finite system over  $\mathbb{Z}$ ,  $V(\mathbf{Oz})$  is nonempty exactly when  $V(\mathbb{Z})$  is nonempty (theorem 5.1). This does not include inequations or order conditions (proposition 5.3).
- (2) **Constant-product rigidity** [01, 02, 05]. A nonzero constant product of normal forms with no negative exponents has only constant factors. At a nonzero constant level, the omnific points of a product of affine linear forms are exactly the integer points plus purely infinite vectors in the real common kernel of the linear parts (theorem 6.2). Binary forms with two distinct projective factors over  $\mathbb{C}$  at a level in  $\mathbb{C}^\times$  have only constant points in the complex nonnegative-support ring (theorem 6.3); Pell equations with  $D \neq 0$  and number-field norm forms are rigid (corollary 6.5 and theorem 6.10).
- (3) **Separated powers and unimodular Fermat** [05]. For real  $a, b, c \neq 0$  and  $m, n \geq 2$ , every nonnegative-support solution of  $ax^m + by^n = c$  is constant (theorem 7.4). For  $n \geq 3$ , a nonzero Fermat triple generating the unit ideal is constant (theorem 7.9); the threshold is sharp.
- (4) **Quadratic-level dichotomy** [01, 02]. For  $q \in \mathbb{Z}[X_1, \dots, X_n]$  quadratic homogeneous and  $0 \neq c \in \mathbb{Z}$ , all solutions are ordinary in the nondegenerate definite case and in nondegenerate dimension at most two. In every other case each integer point lifts to an injectively parametrized infinite family (theorem 8.1). For indefinite forms in at least three variables,  $(\Pi, +)$  embeds in the kernel of  $\text{SO}(q, \mathbf{Oz}) \rightarrow \text{SO}(q, \mathbb{Z})$  (corollary 8.4).

(5) **Quartic definitions of  $\mathbb{Z}$**  [01, 05]. The equation

$$(U^2 - 2V^2 - 1)^2 + \left( U - \sum_{i=1}^n X_i^2 - W_1^2 - W_2^2 - W_3^2 \right)^2 = 0$$

with five existentially quantified auxiliary variables defines exactly  $\mathbb{Z}^n$  inside  $\mathbf{Oz}^n$  ([theorem 10.3](#)); a variant with six auxiliary variables and four squares defines  $\mathbb{Z}$  ([theorem 10.4](#)). The graph of  $\text{ct}$  and the ideal  $\Pi$  are first-order definable, and an explicit existential induction instance fails ([theorem 10.9](#)).

(6) **Primitive and projective phenomena** [01, 02, 05]. There are infinite unimodular Pythagorean triples. A constant real projective direction has a primitive omnific representative precisely when it is rational ([proposition 11.4](#) and [theorem 11.5](#)).

(7) **Bounded and homogeneous geometry** [01, 02]. Bounded semialgebraic sets have only integer omnific points ([theorem 9.2](#)); a homogeneous system has a nonzero omnific point exactly when it has a nonzero real point, and for integer coefficients this is decidable ([theorem 11.2](#)).

These statements are proved here, but proving a statement here is not a claim that nobody has previously proved it. The constant-term formalism is classical, and several consequences are natural applications of it. Each source names its own most distinctive contributions ([appendix B](#)); no exhaustive historical-priority claim is made.

The *universal set-sized quotient theorem*, that every unital ring homomorphism from the full class  $\mathbf{Oz}$  to a set-sized ring is  $x \mapsto \text{ct}(x)1$ , was proved by source 05 and is printed with its proof in the sibling report `docs/surreal/set-sized-quotients-of-omnific-integers`. It is quoted here as [cited theorem 3.6](#), together with the elementary special cases proved by source 01.

## 1.2 Provenance of this report

This report is built from three manuscripts written independently on September 22, 2026, each pinned to repository commit `2cb9c0200af749fbbd27796c97d017e7f16fbf33` (107 commits before the placement commit `be06fc8`).

| Source | Manuscript                                               | Principal contribution to this report                                                                                                                                                                                                                                                                      |
|--------|----------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 01     | <i>Omnific Integers and Omnific-Diophantine Geometry</i> | Base text and structure; exhaustive quadratic-level dichotomy; the five-auxiliary three-square guard; rational-direction theorem; no real point at infinity; symmetric-matrix rigidity; elementary uniqueness of ct.                                                                                       |
| 02     | <i>Omnific Integers and Diophantine Geometry</i>         | Exact decomposable fibers; bounded semialgebraic rigidity; decidable homogeneous existence; quadratic-formula criterion; general pair without a gcd; explicit Lorentz matrix and the unipotent subgroup.                                                                                                   |
| 05     | <i>Omnific Integers and Diophantine Rigidity</i>         | Binary rigidity over $\mathbb{C}$ ; Euler derivations; separated-power and unimodular Fermat rigidity; the four-square definition of $\mathbb{Z}$ ; failed existential induction; failed lifting; finite-support specialization; the universal set-sized quotient theorem (printed in the sibling report). |

Where the sources overlap, the statement with the weakest hypotheses is printed and the others are recorded as special cases. Results added by the merge carry the tag [merge] and have complete proofs. [Appendix A](#) records, section by section, which source proved what, where the merge had to choose, and every symbol renamed from a source.

**Current review and placement status.** Commit `bdd536` assembled sources 01, 02 and 05. The elementary algebra review and its preservation through this merge are recorded in `RECONCILIATION.md`; assembly is not a review of every later proof. Two further Diophantine manuscripts, locally numbered 06 and 07 and placed in `cf350b1`, are not yet integrated. Nine earlier labels renamed or combined during assembly remain as aliases, so existing references resolve.

### 1.3 Class size and foundations

The full surreal field and  $\mathbf{Oz}$  are proper classes. We work in a customary class theory such as NBG, with the usual background for surreal normal forms; source 02 notes that one may equally read everything relative to universes, with “set-sized” replaced by “small in the specified lower universe”. Each individual surreal has a *set-sized* support. Every polynomial, system, matrix, or tuple in this report has ordinary finite size. Quantifiers over  $\mathbf{Oz}$  are class quantifiers expressed through their defining set-theoretic formulas, not an appeal to a set containing all surreals.

Quotients by class ideals are interpreted through their quotient maps and congruence relations. Thus  $\mathbf{Oz}/\Pi \cong \mathbb{Z}$  uses the ordinary integer  $\text{ct}(x)$  as representative; for  $0 \neq n \in \mathbb{Z}$ ,  $\mathbf{Oz}/n\mathbf{Oz}$  uses the ordinary residue of  $\text{ct}(x)$  modulo  $n$ . Proper-class cosets are not elements of a set. Ideal generation uses finite sums, even for a class of generators. The notation  $\text{Frac}(\mathbf{Oz}) = \mathbf{No}$  means representation by fractions inside  $\mathbf{No}$ , not a set of all pairs from  $\mathbf{Oz}$ .

Choice for sets is used to extend a rational linear functional on a set-sized exponent group

([lemma 7.2](#)). No proof printed in this report uses global choice. Source 05 works in NBG with global choice, which it uses only to pick a cardinal in the proof of the set-sized quotient theorem; the Hartogs ordinal of the target ring serves there without choice, since the pigeonhole step needs only that no injection from that ordinal into the target exists ([remark 3.7](#)).

Most proofs can be carried out in a fixed set-sized Hahn field. We identify the arguments that need the stronger fact that *every set* of surreal exponents has upper bounds and every set of positive surreal exponents has a strictly positive lower bound ([section 4](#)). One must not silently assume these properties for a fixed ordered exponent group ([section 13](#)).

## 1.4 Notation and the exponent convention

| Notation                                                                                                         | Meaning in this report                                                                                                        |
|------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------|
| $\omega^\gamma$                                                                                                  | Conway's monomial; $\omega^\gamma \omega^\delta = \omega^{\gamma+\delta}$ with surreal addition of exponents.                 |
| $\Pi$                                                                                                            | The ideal of normal forms supported on $\mathbf{No}^{>0}$ (purely infinite elements and 0).                                   |
| $\mathbf{Oz} = \mathbb{Z} \oplus \Pi$                                                                            | Omnific integers: nonnegative support, integer constant coefficient.                                                          |
| $\mathcal{B}_{\mathbb{R}} = \mathbb{R} \oplus \Pi$ ,<br>$\mathcal{B}_{\mathbb{C}} = \mathcal{B}_{\mathbb{R}}[i]$ | Nonnegative-support rings with real or complex coefficients;<br>$\Pi_{\mathbb{C}} = \Pi + i\Pi$ .                             |
| $\mathcal{B}_{\mathbb{R},H}, \mathcal{B}_{\mathbb{C},H}$                                                         | Their subrings with exponents in a set-sized divisible group $H \subseteq \mathbf{No}$ ( <a href="#">lemma 2.6</a> ).         |
| $\text{ct}(x)$                                                                                                   | Coefficient of $\omega^0$ ; not an order standard part.                                                                       |
| $\deg_\omega(x), \text{lc}(x)$                                                                                   | Largest exponent in $\text{supp}(x)$ and its coefficient;<br>$\deg_\omega(0) = -\infty$ . Not an ordinal and not a birthday.  |
| $\text{Std}(t), \text{Std}_n(\mathbf{x})$                                                                        | The existential formulas of <a href="#">theorems 10.3</a> and <a href="#">10.4</a> defining $\mathbb{Z}$ and $\mathbb{Z}^n$ . |
| $\partial_\lambda, W_\partial$                                                                                   | Local Euler derivation and two-term Wronskian ( <a href="#">section 7</a> ).                                                  |
| $B, q, G$                                                                                                        | Symmetric bilinear form, its quadratic form $q(x) = B(x, x)$ , and the Lorentz Gram matrix $G = \text{diag}(1, -1, -1)$ .     |
| $R, S$                                                                                                           | A generic commutative ring; a set-sized target ring.                                                                          |

The calligraphic  $\mathcal{B}_{\mathbb{R}}, \mathcal{B}_{\mathbb{C}}$  name rings; the italic  $B$  is always a bilinear form. The capital Greek letter  $\Pi$  always denotes the ideal; finite products are always written with the operator  $\prod$  and its index range.

**Sign convention.** Exponents are those of Conway normal forms: supports are reverse well ordered, and  $\omega^\gamma$  is *infinite* for  $\gamma > 0$ , even for infinitesimal  $\gamma > 0$ . Thus

$$\Pi = \{x \in \mathbf{No} : \text{supp}(x) \subseteq \mathbf{No}^{>0}\} \quad (\omega\text{-exponents}).$$

In the Hahn convention  $t = \omega^{-1}$ , which is the convention of the trigonometry report ([docs/surcomplex/trigonometry](#), equation `trigonometry:eq:split`), every exponent changes sign:  $\omega^\gamma = t^{-\gamma}$ , supports become well ordered, the valuation is  $v = -\deg_\omega$ , and the same class is

$$\Pi = \{x \in \mathbf{No} : \text{supp}_t(x) \subseteq \mathbf{No}^{<0}\} \quad (t\text{-exponents}).$$

So “strictly positive exponents” here and “negative  $t$ -exponents” there describe one class, and “no positive exponents” in the  $t$ -convention is “no negative exponents” here. The restriction

concerns every supported exponent, not only the leading one:  $\mathcal{B}_{\mathbb{R}}$  consists of the forms whose entire  $t$ -support is nonpositive. It is not the set of elements with nonpositive valuation; for example  $\omega + \omega^{-1}$  has negative valuation but does not lie in  $\mathcal{B}_{\mathbb{R}}$ . Nor is  $\mathcal{B}_{\mathbb{R}}$  the valuation ring of finite elements. Using the wrong convention would invalidate the unit computation (proposition 2.5) and every rigidity proof below.

The repository’s foundations report defines  $\mathbf{Oz} = \Pi \oplus \mathbb{Z}$  with the same letter  $\Pi$  (docs/foundations-and-computation/foundations, label found:eq:omnific); the notation here agrees with it. Other repository names for the same class are  $\mathbf{No}^{\uparrow}$  and  $\mathbf{No}_{\text{PI}}$  in other reports. The sources used  $\mathcal{J}$  (01) and  $\mathcal{I}$  (02, 05); the complete per-source rename table is table 1.

## 1.5 Relationship with the repository and literature

The repository distinguishes normal-form algebra, fixed Hahn workspaces, and the full surreal carrier. At the sources’ pin, the foundations report already defined  $\mathbf{Oz} = \Pi \oplus \mathbb{Z}$  (found:eq:omnific), and the trigonometry report already used the omnific integer part, the floor formula and residue arithmetic (trigonometry:eq:periodclass); source 01 mentioned only the latter. At the pin there was no report devoted to  $\mathbf{Oz}$  and no Lean declaration about it. The collection now contains this report and the sibling report docs/surreal/set-sized-quotients-of-omnific-integers; there is still no Lean formalization of omnific integers (section 13 and appendix D).

The main external sources are Conway and Gonshor for the construction and normal forms [1, 2], L’Innocente–Mantova for modern factorization theory [3], Ehrlich–Kaplan for the initial integer-part context [4], and Jeřábek for the residue theory of  $\mathbb{Z}$ -groups [5]. MRDP, Tarski’s quantifier elimination, the three-square and four-square theorems, Shepherdson’s theorem and the Fermat theorem of Wiles and Taylor–Wiles are imported with explicit attribution at their points of use. Relations to neighbouring reports are listed in section 14.4.

## 2 The normal-form rings and the integer part

### 2.1 Normal forms and three related rings

We use the standard normal-form theorem: each surreal has a unique expression

$$x = \sum_{\gamma \in S} r_{\gamma} \omega^{\gamma}, \quad r_{\gamma} \in \mathbb{R} \setminus \{0\}, \quad (2.1)$$

where  $S \subseteq \mathbf{No}$  is a reverse-well-ordered set: every nonempty subset has a greatest member. Conversely such a normal form represents a surreal. Addition is coefficientwise, multiplication is Hahn convolution (the sum of two reverse-well-ordered supports is reverse well ordered, and each exponent of a product receives finitely many contributions), and the sign of a nonzero normal form is the sign of its leading coefficient [1, 2]. The field  $\mathbf{No}$  is real closed.

**Definition 2.1.** Let

$$\begin{aligned} \Pi &= \{x \in \mathbf{No} : \text{supp}(x) \subseteq \mathbf{No}^{>0}\}, \\ \mathcal{B}_{\mathbb{R}} &= \mathbb{R} \oplus \Pi, \\ \mathbf{Oz} &= \mathbb{Z} \oplus \Pi, \\ \mathcal{B}_{\mathbb{C}} &= \mathcal{B}_{\mathbb{R}}[i] \subseteq \mathbf{No}[i]. \end{aligned}$$

The zero series belongs to  $\Pi$ . For an element of any of these rings,  $\text{ct}(x)$  denotes its coefficient at exponent zero. For  $x = a + ib \in \mathcal{B}_{\mathbb{C}}$ , this is  $\text{ct}(a) + i \text{ct}(b)$ . Put  $\Pi_{\mathbb{C}} = \Pi + i\Pi$ , so  $\mathcal{B}_{\mathbb{C}} = \mathbb{C} \oplus \Pi_{\mathbb{C}}$ .

Thus  $\sqrt{2}\omega$ ,  $\omega/17$ ,  $-\omega^\omega + 3$ ,  $\omega^{1/\omega}$  and  $\sum_{k \geq 1} \omega^{1/k}$  are omnific integers. The numbers  $\omega + \frac{1}{2}$  and  $\omega + \omega^{-1}$  are not. The word “integer” restricts the constant coefficient and excludes infinitesimal tails; it does not require all coefficients to lie in  $\mathbb{Z}$ . “Purely infinite” is a support condition, not a sign condition:  $-\omega \in \Pi$ , while  $\omega - 3 \notin \Pi$ . The normal-form definition of  $\mathbf{Oz}$  is equivalent to Conway’s simplicity characterization by the cut  $\{x - 1 \mid x + 1\}$  [1, 2]; no proof below uses that characterization.

For every positive surreal exponent  $\gamma$ , even an infinitesimal one, the Conway monomial  $\omega^\gamma$  exceeds every ordinary positive real number. This is a normal-form order fact about the omega-map; replacing that map by an unstated analytic expression  $\exp(\gamma \log \omega)$  would invalidate the support arguments below.

**Proposition 2.2** (Rings and retraction [01, 02, 05]).  *$\mathcal{B}_\mathbb{R}$ ,  $\mathcal{B}_\mathbb{C}$ , and  $\mathbf{Oz}$  are integral domains, and  $\mathcal{B}_\mathbb{R}$ ,  $\mathbf{Oz}$  are ordered subrings of  $\mathbf{No}$ . The class  $\Pi$  is an ideal of  $\mathcal{B}_\mathbb{R}$  and  $\mathbf{Oz}$ , and is an  $\mathbb{R}$ -vector space;  $\Pi_\mathbb{C}$  is an ideal of  $\mathcal{B}_\mathbb{C}$ . Constant-coefficient extraction is a surjective ring homomorphism from  $\mathcal{B}_\mathbb{R}$  to  $\mathbb{R}$ , from  $\mathcal{B}_\mathbb{C}$  to  $\mathbb{C}$ , and from  $\mathbf{Oz}$  to  $\mathbb{Z}$ . In the last case its kernel is  $\Pi$ , the inclusion  $\mathbb{Z} \hookrightarrow \mathbf{Oz}$  is a section, and  $\mathbf{Oz}/\Pi \cong \mathbb{Z}$ .*

*Proof.* Nonnegative exponents remain nonnegative under addition of exponents. A strictly positive exponent plus a nonnegative exponent is strictly positive. These observations and Hahn convolution prove the ring and ideal claims. In a product of two supported normal forms, exponent zero can arise only as  $0 + 0$ . Therefore  $\text{ct}(xy) = \text{ct}(x)\text{ct}(y)$ , and the constant coefficient of a product of elements of  $\mathbf{Oz}$  is an integer. Additivity is coefficientwise. Real scalar multiplication does not change a support except when the scalar is zero. Surjectivity and the section are supplied by constant series. All rings embed in a field, so have no zero divisors.  $\square$

*Remark 2.3* (Two misleading analogies). The map  $\text{ct}$  is not a ring homomorphism on all of  $\mathbf{No}$ :  $\text{ct}(\omega) = \text{ct}(\omega^{-1}) = 0$  but  $\text{ct}(1) = 1$ . The repository records the same warning for coefficient extraction at zero on  $\mathbf{No}$  (`diff:warn:ct` in the differential-equations report, and `docs/NOTATION.md`); the map  $\text{ct}$  of this report is the restriction of that extraction to  $\mathbf{Oz}$ , where it *is* multiplicative. Also  $\mathcal{B}_\mathbb{R}$  is not the ring of finite surreals. The latter permits negative exponents but no positive exponents;  $\mathcal{B}_\mathbb{R}$  has the opposite support restriction. Taking the standard part of a finite element and taking the constant coefficient of an omnific integer are different constructions, although both are multiplicative on their own domains. Finally,  $\text{ct}$  is not order preserving:  $\omega - 5 > 0$  but  $\text{ct}(\omega - 5) = -5$ .

## 2.2 Leading degree, units, and finite elements

For nonzero  $x \in \mathbf{No}[i]$ , combine real and imaginary coefficients and put

$$\deg_\omega(x) = \max \text{supp}(x), \quad \text{lc}(x) = [\omega^{\deg_\omega(x)}]x, \quad \deg_\omega(0) = -\infty.$$

On  $\mathcal{B}_\mathbb{C} \setminus \{0\}$  the degree lies in  $\mathbf{No}^{\geq 0}$ .

**Lemma 2.4** (Degree [01, 02, 05]). *For nonzero normal forms  $x, y$  with real or complex coefficients,*

$$\begin{aligned} \deg_\omega(xy) &= \deg_\omega(x) + \deg_\omega(y), & \text{lc}(xy) &= \text{lc}(x)\text{lc}(y), \\ \deg_\omega(x + y) &\leq \max\{\deg_\omega x, \deg_\omega y\}, \end{aligned}$$

*with equality in the last relation when  $\deg_\omega x \neq \deg_\omega y$ . An element of  $\mathcal{B}_\mathbb{C}$  of degree zero is a nonzero complex constant.*

*Proof.* If  $\gamma$  and  $\delta$  are the respective largest exponents, every contributing exponent of  $xy$  is at most  $\gamma + \delta$ . Equality requires both exponents to be largest: making either smaller makes their sum smaller. Thus the coefficient at  $\gamma + \delta$  is the nonzero product of leading coefficients. The statements about sums follow from cancellation at the greatest exponent. Finally, support in  $[0, 0]$  contains only zero.  $\square$

**Proposition 2.5** (Units and finite elements [01, 02, 05]). *The unit groups are*

$$\mathcal{B}_{\mathbb{R}}^{\times} = \mathbb{R}^{\times}, \quad \mathcal{B}_{\mathbb{C}}^{\times} = \mathbb{C}^{\times}, \quad \mathbf{Oz}^{\times} = \{1, -1\}.$$

*The omnific integers bounded in absolute value by some ordinary real number are exactly the ordinary integers; in particular  $\mathbf{Oz} \cap \mathbb{R} = \mathbb{Z}$ . Consequently  $\mathbf{Oz}$  is discretely ordered, with least positive element 1.*

*Proof.* If  $xy = 1$  in  $\mathcal{B}_{\mathbb{C}}$ , nonnegativity and additivity of degree force both degrees to be zero. This gives the first two unit statements; in  $\mathbf{Oz}$  the constant units are precisely the units of  $\mathbb{Z}$ .

A nonconstant element of  $\mathbf{Oz}$  has positive leading exponent, so its absolute value exceeds every positive real number by normal-form order. A constant element of  $\mathbf{Oz}$  is an integer. Hence no element lies strictly between 0 and 1, and every positive element is at least 1.  $\square$

Notice that  $\omega/2$  is an omnific integer even though  $1/2$  is not. This is no conflict with discreteness:  $\omega/2$  is infinite, and there are still no omnific integers in the interval  $(0, 1)$ .

The unit computation is stronger than saying that an infinite factor has large real absolute value: complex leading coefficients can cancel in sums, but they cannot cancel the unique leading coefficient in a product. It will do much of the Diophantine work.

### 2.3 Local workspaces

**Lemma 2.6** (Set-sized workspaces [05]). *Any set-sized family of normal forms belongs to a common Hahn field whose exponents lie in a set-sized ordered divisible group  $H \subseteq \mathbf{No}$ . One can take  $H$  to be the rational span of the union of the supports.*

*Proof.* The union of the supports is a set, and so is its rational span. Normal forms supported in  $H$  constitute the real or complex Hahn field with exponents in  $H$ . The Hahn field operations preserve this exponent group, including inverses of nonzero series.  $\square$

We write  $\mathcal{B}_{\mathbb{R},H}$ ,  $\mathcal{B}_{\mathbb{C},H}$  and  $\mathbf{Oz}_H$  for the intersections of  $\mathcal{B}_{\mathbb{R}}$ ,  $\mathcal{B}_{\mathbb{C}}$  and  $\mathbf{Oz}$ , respectively, with the workspace; in particular  $\mathbf{Oz}_H$  retains the integer constant-coefficient condition. Most proofs below, including all the main equation-rigidity theorems, take place in one such set-sized ring and need no proper-class derivation or basis. By contrast, a positive surreal smaller than every member of a specified set of positive exponents need not belong to a preassigned  $H$ ; the results of [section 4](#) genuinely use the full exponent class, and are marked there.

### 2.4 The exact omnific floor

Every surreal admits a unique decomposition

$$x = x_{>0} + r + \varepsilon, \quad x_{>0} \in \Pi, \quad r \in \mathbb{R}, \quad \text{supp}(\varepsilon) \subseteq \mathbf{No}^{<0}.$$

Here  $\varepsilon$  is infinitesimal, possibly zero.

**Theorem 2.7** (Integer-part formula [01, 02, 05]). *There is exactly one  $a \in \mathbf{Oz}$  such that  $a \leq x < a + 1$ . It is*

$$\lfloor x \rfloor_{\mathbf{Oz}} = \begin{cases} x_{>0} + \lfloor r \rfloor, & r \notin \mathbb{Z} \text{ or } \varepsilon \geq 0, \\ x_{>0} + r - 1, & r \in \mathbb{Z} \text{ and } \varepsilon < 0. \end{cases} \quad (2.2)$$

*Proof.* For  $r \notin \mathbb{Z}$ , its distances from the two adjacent integers are strictly positive real numbers, and the infinitesimal  $\varepsilon$  cannot cross either boundary. If  $r$  is an integer and  $\varepsilon \geq 0$ , the remainder is  $\varepsilon \in [0, 1)$ . If  $r$  is an integer and  $\varepsilon < 0$ , the remainder after subtracting  $x_{>0} + r - 1$  is  $1 + \varepsilon \in (0, 1)$ . Two possible floors would differ by an omnific integer strictly between  $-1$  and  $1$ , so would coincide by discreteness.  $\square$

For example,

$$\lfloor \omega - \omega^{-1} \rfloor_{\mathbf{Oz}} = \omega - 1, \quad \lfloor \omega + \frac{3}{2} - \omega^{-1} \rfloor_{\mathbf{Oz}} = \omega + 1, \quad \lfloor -\omega^{-1} \rfloor_{\mathbf{Oz}} = -1.$$

Merely discarding negative exponents and flooring the constant gives the wrong answer at a negative infinitesimal displacement from an integer.

*Provenance.* The integer-part assertion is classical. Ehrlich–Kaplan prove the stronger initiality characterization: for an initial subfield  $K$  of  $\mathbf{No}$ ,  $\mathbf{Oz} \cap K$  is its unique initial integer part [4, Lemma 11.1]. No alternative integer part is introduced. The formula agrees with the trigonometry report’s construction of the integer part, which takes “the appropriate side when  $f$  is infinitesimally adjacent to an integer” (text following `trigonometry:eq:split`).

**Proposition 2.8** (One-variable rigidity [02, 05]). *If  $P \in \mathbb{R}[T]$  is nonzero, every root of  $P$  in  $\mathcal{B}_{\mathbb{R}}$  belongs to  $\mathbb{R}$ . In particular, the roots in  $\mathbf{Oz}$  of a nonzero  $P \in \mathbb{Z}[T]$  are exactly its ordinary integer roots.*

*Proof.* If a root  $x$  were nonconstant,  $\alpha = \deg_{\omega}(x) > 0$ . For nonzero coefficients  $a_j$ , the terms  $a_j x^j$  have distinct degrees  $j\alpha$ , so the highest-degree term cannot cancel. Thus  $P(x) \neq 0$ . A constant root in  $\mathbf{Oz}$  is an integer.  $\square$

Every infinite omnific integer is therefore transcendental over  $\mathbb{R}$ . This does not make a tuple of infinite omnific integers algebraically independent:  $(\omega, \omega^2)$  is a counterexample. Infinite coefficients change the problem:  $X^2 - \omega^2$  has the infinite roots  $\pm\omega$ .

### 3 Divisibility, residues, and quotients

#### 3.1 Division by ordinary integers

**Theorem 3.1** (Ordinary moduli [01, 02, 05]). *For every nonzero ordinary integer  $n$  and  $x \in \mathbf{Oz}$ ,*

$$n\mathbf{Oz} = \Pi + n\mathbb{Z}, \quad n \mid x \iff n \mid \text{ct}(x), \quad \mathbf{Oz}/n\mathbf{Oz} \cong \mathbb{Z}/n\mathbb{Z}. \quad (3.1)$$

*In particular, for every ordinary prime  $p$ ,*

$$\bigcap_{n \geq 1} n\mathbf{Oz} = \Pi = \bigcap_{k \geq 1} p^k \mathbf{Oz}. \quad (3.2)$$

*If  $\text{ct}(x) \neq 0$ , the largest ordinary  $k$  with  $p^k \mid x$  is the  $p$ -adic order of  $\text{ct}(x)$ ; if  $\text{ct}(x) = 0$ , every power of every ordinary prime divides  $x$ .*

*Proof.* Necessity in the divisibility criterion follows by applying  $\text{ct}$  to  $x = ny$ . Multiplication by a nonzero real scalar preserves  $\Pi$  and is invertible on  $\Pi$ , so  $n\Pi = \Pi$ . If  $x = m + j$  with  $m \in \mathbb{Z}$ ,  $j \in \Pi$ , then  $x/n = m/n + j/n$  lies in  $\mathbf{Oz}$  exactly when  $m/n$  is an integer. Reduction of  $\text{ct}$  modulo  $n$  is therefore a surjection with kernel  $n\mathbf{Oz}$ . The only ordinary integer divisible by all positive integers, or by every power of one prime, is zero, which gives (3.2) and the statement on  $p$ -adic orders.  $\square$

This theorem solves every ordinary congruence. For instance,  $\sqrt{2}\omega + 14$  is even and is congruent to 2 modulo 3;  $\omega$  is divisible by every positive ordinary integer and  $\omega + 1$  is odd. The whole infinite part is invisible to these tests.

The statement does *not* extend to arbitrary omnific divisors. For example,  $\text{ct}(\omega + 2)$  divides  $\text{ct}(2)$  in  $\mathbb{Z}$ , but  $\omega + 2$  does not divide 2 in  $\mathbf{Oz}$ . More generally:

**Proposition 3.2** (Divisors of integers [05]). *If  $d \in \mathbf{Oz}$  divides a nonzero ordinary integer  $N$ , then  $d$  is an ordinary integer divisor of  $N$ .*

*Proof.* Write  $de = N$ . Neither factor is zero, so  $|e| \geq 1$  by discreteness and  $|d| \leq |N|$ . Hence  $d \in \mathbb{Z}$  by proposition 2.5; similarly  $e \in \mathbb{Z}$ .  $\square$

**Corollary 3.3** (Ordinary primes and mixed gcds [02, 05]). *Every positive ordinary prime  $p$  is a prime element of  $\mathbf{Oz}$ , and  $p\mathbf{Oz}$  is maximal. For  $0 \neq n \in \mathbb{Z}$  and  $x \in \mathbf{Oz}$ ,*

$$(n, x) = \gcd(n, \text{ct}(x)) \mathbf{Oz},$$

*so an ordinary gcd of  $n$  and  $\text{ct}(x)$  is a gcd of  $n$  and  $x$  in  $\mathbf{Oz}$ . The Chinese remainder theorem for pairwise coprime ordinary moduli holds in  $\mathbf{Oz}$  with exactly the ordinary residue rings.*

*Proof.* The quotient by  $p\mathbf{Oz}$  is the field  $\mathbb{Z}/p\mathbb{Z}$ . If  $x = m + j$  with  $j \in \Pi$ , then  $x - n(j/n) = m$ , so  $(n, x) = (n, m)$ , which an ordinary Bézout identity identifies. A generator of a two-generated principal ideal is a gcd of its generators in any domain. For coprime moduli, ordinary Bézout coefficients give comaximality, and the quotient descriptions give the Chinese remainder isomorphism.  $\square$

**Corollary 3.4** (Positive-characteristic quotients [01, 02, 05]). *Every proper ideal  $\mathfrak{a} \subset \mathbf{Oz}$  containing a nonzero ordinary integer is  $n\mathbf{Oz}$  for a unique ordinary  $n \geq 2$ ; it is prime exactly when  $n$  is an ordinary prime. Every unital homomorphism from  $\mathbf{Oz}$  to a finite ring kills  $\Pi$  and factors through  $\text{ct}$ , and every nonzero finite quotient of  $\mathbf{Oz}$  is a ring  $\mathbb{Z}/n\mathbb{Z}$ .*

*Proof.* If  $m \in \mathfrak{a}$  with  $0 \neq m \in \mathbb{Z}$ , then  $\Pi = m\Pi \subseteq \mathfrak{a}$ . Thus  $\mathfrak{a}$  is the inverse image of an ideal of  $\mathbf{Oz}/\Pi = \mathbb{Z}$ , namely  $n\mathbb{Z}$  with  $n > 0$ , and properness gives  $n \geq 2$ . Equation (3.1) identifies its inverse image with  $n\mathbf{Oz}$ ; primality is that of  $n\mathbb{Z}$ . If a finite target has characteristic  $m$ , then each  $j \in \Pi$  is  $m(j/m)$  and maps to zero. A finite nonzero unital ring has positive characteristic, so the last assertion follows.  $\square$

## 3.2 Completions

All finite-index ideals have just been classified, so the profinite completion, interpreted as the set-sized inverse limit over ordinary  $n \geq 1$ , is

$$\widehat{\mathbf{Oz}} \cong \varprojlim_n \mathbb{Z}/n\mathbb{Z} = \widehat{\mathbb{Z}}, \quad \varprojlim_k \mathbf{Oz}/p^k \mathbf{Oz} \cong \mathbb{Z}_p. \quad (3.3)$$

The completion maps are the composites of  $\text{ct}$  with the canonical maps of  $\mathbb{Z}$ , with kernel  $\Pi$  in both cases by (3.2). Thus these congruence topologies on  $\mathbf{Oz}$  are not Hausdorff and  $\mathbf{Oz}$  is not residually finite [01, 02, 05]. For example  $\omega$ ,  $\sqrt{2}\omega$  and 0 have identical images in every finite quotient. The separated quotient is  $\mathbf{Oz}/\Pi = \mathbb{Z}$ ; completion does not preserve the infinite part.

The first inverse system is indexed by positive integers ordered by divisibility, with reduction maps  $\mathbb{Z}/n\mathbb{Z} \rightarrow \mathbb{Z}/m\mathbb{Z}$  for  $m \mid n$ . The second uses the powers of a fixed ordinary prime  $p$ . These are set-sized diagrams of ordinary residue rings, not sets of class ideals or proper-class cosets.

### 3.3 Homomorphisms to $\mathbb{Z}$ and to set-sized rings

**Proposition 3.5** (Uniqueness of the retraction [01]). *The constant-term map is the unique unital ring homomorphism  $\mathbf{Oz} \rightarrow \mathbb{Z}$ . Every unital endomorphism  $\psi : \mathbf{Oz} \rightarrow \mathbf{Oz}$  preserves  $\Pi$  and satisfies  $\text{ct} \circ \psi = \text{ct}$ .*

*Proof.* For a homomorphism  $h : \mathbf{Oz} \rightarrow \mathbb{Z}$ , the integer  $h(j)$  is divisible by every positive integer whenever  $j \in \Pi$ . Hence  $h(j) = 0$ . Since  $h$  fixes ordinary integers,  $h = \text{ct}$ . An endomorphism preserves divisibility by each ordinary integer, so preserves their common divisibility kernel (3.2). It fixes  $\mathbb{Z}$  pointwise, which proves the final identity.  $\square$

This does not say that every endomorphism is the identity: the composite of  $\text{ct}$  with the inclusion  $\mathbb{Z} \hookrightarrow \mathbf{Oz}$  is a noninjective unital endomorphism [05].

*Provenance.* The divisible-kernel and residue-map perspective belongs to classical  $\mathbb{Z}$ -group theory; see [5, Sections 2 and 4]. Here it has a concrete ring-theoretic realization because  $\text{ct}$  is multiplicative. The finite-quotient and completion statements above are proved directly.

Source 05 proved a far-reaching generalization of [proposition 3.5](#) and [corollary 3.4](#). The sibling report gives a more general coefficient-ring theorem in its installed base manuscript; its companion-source integration remains pending. For that proof see the sibling report `docs/surreal/set-sized-quotients-of-omnific-integers`; we quote it without proof.

**Cited theorem 3.6** (Universal set-sized quotient; proved by source 05, printed in the sibling report). *Let  $S$  be a set-sized unital ring, not necessarily commutative, and let  $\varphi : \mathbf{Oz} \rightarrow S$  be a unital class ring homomorphism. Then  $\varphi(x) = \text{ct}(x) 1_S$  for every  $x \in \mathbf{Oz}$ . Consequently every set-sized quotient ring of  $\mathbf{Oz}$  is isomorphic to  $\mathbb{Z}$  (kernel  $\Pi$ ) or to  $\mathbb{Z}/n\mathbb{Z}$  (kernel  $n\mathbf{Oz}$ ) for an ordinary  $n \geq 1$ .*

Source 05's proof applies pigeonhole to the monomials  $\omega^{\delta/\omega^{\alpha+1}}$  for ordinals  $\alpha$  beyond the size of  $S$ , and kills a positive monomial  $\omega^\delta$  by the exact Hahn identity  $(\omega^a - \omega^b) \sum_{k \geq 0} \omega^{\delta-a-k(a-b)} = \omega^\delta$ , whose quotient lies in  $\Pi$  when  $0 < b < a \leq \delta/\omega$ ; it then uses [theorem 4.2](#) below to kill all of  $\Pi$ .

*Remark 3.7* (Foundations of the cited theorem). Source 05 chooses an infinite cardinal  $\kappa > |S|$ , using the global choice of its NBG setting. The Hartogs ordinal of  $S$ , the least ordinal admitting no injection into  $S$ , can be used instead: the pigeonhole step needs only that the family  $(\omega^{\delta/\omega^{\alpha+1}})_\alpha$  indexed by that ordinal does not inject into  $S$ . So neither global choice nor choice for sets is needed for [cited theorem 3.6](#).

[Proposition 3.5](#) is the case  $S = \mathbb{Z}$  of [cited theorem 3.6](#) together with its consequence for endomorphisms (apply it to  $\text{ct} \circ \psi$ ), and [corollary 3.4](#) is the case of finite  $S$ ; the elementary divisibility proofs above are independent second routes. The theorem does not apply to class-sized targets such as  $\mathbf{Oz}$  itself, and it is a full-class statement: the subring  $\mathbb{Z} + \omega\mathbb{R}[\omega]$  of  $\mathbf{Oz}$  admits evaluation at many real values, and such maps do not extend to  $\mathbf{Oz}$  [05]. Source

05 also records that, by [cited theorem 3.6](#), the quotient  $\mathbf{Oz}/(f)$  by an infinite prime  $f$  with  $\text{ct}(f) = \pm 1$  (for example the prime of [\[3, Theorem B\]](#) in [section 4.2](#)) is not set-sized; that consequence is likewise printed in the sibling report.

## 4 Full-class support: common divisors, clearing, factorization

The next results use all surreal exponents, not merely the exponents in one fixed Archimedean or finite-rank group. This is an important source of arithmetic differences between full omnific integers and restricted series rings ([section 13](#)).

### 4.1 One monomial divides a whole set

**Lemma 4.1** (Set-sized exponent bounds [01, 02, 05]). *If  $E \subset \mathbf{No}^{>0}$  is a set, there is a surreal  $\delta > 0$  such that  $\delta < e$  for every  $e \in E$ . Every set of surreals also has a strict upper bound.*

*Proof.* A surreal separating the cut  $\{0 \mid E\}$  gives the first assertion, and a separator of  $\{E \mid \emptyset\}$  the second; their option collections are sets. The assertions also hold for the empty set. These are applications of the set-cut property, not an assertion of an infimum or supremum. No single  $\delta$  works for the entire proper class  $\mathbf{No}^{>0}$ .  $\square$

**Theorem 4.2** (Common monomial divisibility [01, 02, 05]). *For every set  $S \subseteq \Pi$  there is  $\delta > 0$  such that*

$$\omega^\delta \mid s \quad \text{and} \quad s/\omega^\delta \in \Pi \quad (s \in S).$$

*In particular, each nonzero element of  $\Pi$  is a product of two nonzero elements of  $\Pi$ , both nonunits of  $\mathbf{Oz}$ .*

*Proof.* The union of the supports of the elements of  $S$  is a set  $E$  of strictly positive surreal exponents. Choose  $\delta$  as in [lemma 4.1](#). Dividing by  $\omega^\delta$  shifts every exponent  $e$  to  $e - \delta > 0$ , preserving the reverse well-order of each support. The quotient is therefore in  $\Pi$ . For a nonzero  $s$ , both factors are nonzero elements of positive degree, and neither is a unit.  $\square$

Even the series  $\sum_{k \geq 1} \omega^{1/k}$  has such a divisor. The positive exponent  $\delta$  may be infinitesimal relative to every real  $1/k$ , for example  $\delta = 1/\omega$ . This is why the proof cannot be copied into a Hahn ring with exponent group  $\mathbb{R}$ .

**Corollary 4.3** (The purely infinite ideal [01, 02, 05]). *The ideal  $\Pi$  satisfies  $\Pi^2 = \Pi$ , and each element of  $\Pi$  is a single product of two elements of  $\Pi$ . It is not generated as an ideal by any set of elements; in particular it is not finitely generated. Consequently  $\Pi/\Pi^2 = 0$  although  $\Pi \neq 0$ ,  $\mathbf{Oz}/\Pi^k \cong \mathbb{Z}$  for every ordinary  $k \geq 1$ , and the  $\Pi$ -adic completion of  $\mathbf{Oz}$  is  $\mathbb{Z}$ . No nonzero element of  $\Pi$  is a finite product of irreducibles of  $\mathbf{Oz}$ .*

*Proof.* The inclusion  $\Pi^2 \subseteq \Pi$  follows from the ideal property. For  $j \neq 0$ , the factorization in [theorem 4.2](#) has both factors in  $\Pi$ , proving the reverse inclusion. Hence  $\Pi^k = \Pi$  for all  $k \geq 1$  and the inverse system  $\mathbf{Oz}/\Pi^k$  is constantly  $\mathbb{Z}$ .

Suppose a set  $S \subseteq \Pi$  generated  $\Pi$ . Choose  $\delta > 0$  strictly below every exponent in the supports of its elements. Each finite  $\mathbf{Oz}$ -linear combination of elements of  $S$  has support strictly above  $\delta$ : multiplication by an element of  $\mathbf{Oz}$  only adds nonnegative exponents. But  $\omega^{\delta/2} \in \Pi$  does not have such a support.

Finally, if  $j = p_1 \cdots p_k \neq 0$  with every  $p_i$  irreducible, then  $0 = \text{ct}(j) = \prod_{i=1}^k \text{ct}(p_i)$  would put some  $p_i$  in  $\Pi$ , and that factor would be reducible by [theorem 4.2](#).  $\square$

The factorization  $j = 2(j/2)$  already proves that a nonzero  $j \in \Pi$  is reducible, even in a fixed exponent workspace; the stronger content of [theorem 4.2](#) is a common *infinite monomial* divisor for an entire set, with all quotients still in  $\Pi$ . The vanishing of  $\Pi/\Pi^2$  does not say that the infinite part is small; it says that this particular quotient cannot detect it. The hypotheses of a finitely generated Nakayama argument are not present [02].

**Corollary 4.4** (Nilpotent tests [02]). *Let  $\phi : \mathbf{Oz} \rightarrow R$  be a ring homomorphism and  $\mathfrak{n} \subseteq R$  a nilpotent ideal with  $\phi(\Pi) \subseteq \mathfrak{n}$ . Then  $\phi(\Pi) = 0$ .*

*Proof.* Choose an ordinary  $N \geq 1$  with  $\mathfrak{n}^N = 0$ . By [corollary 4.3](#),  $\Pi^N = \Pi$ . The image of every finite sum of  $N$ -fold products from  $\Pi$  belongs to  $\mathfrak{n}^N = 0$ . This proves the claim without assuming that  $\phi(\Pi)$  is an ideal of all of  $R$ .  $\square$

## 4.2 Factorization

[Corollary 4.3](#) is an explicit failure of atomic factorization, not merely a failure of uniqueness of factorization. It leaves room for nonconstant prime elements having nonzero constant coefficient.

**Proposition 4.5** (Irreducibles [01, 02, 05]). *The constant irreducibles of  $\mathbf{Oz}$  are exactly the ordinary signed primes. Every nonconstant irreducible has constant coefficient 1 or  $-1$ . The latter condition is not sufficient for irreducibility. In particular,  $\mathbf{Oz}$  is neither atomic nor a unique factorization domain.*

*Proof.* Constant irreducibles are classified by [proposition 3.2](#). If a nonconstant element  $a$  has  $|\text{ct}(a)| > 1$ , choose an ordinary prime  $p$  dividing its constant coefficient. Then  $a = p(a/p)$  by [theorem 3.1](#) is a factorization into two nonunits: the second factor remains nonconstant. If  $\text{ct}(a) = 0$ , the factorization  $a = 2(a/2)$  has the same property. This leaves only constant coefficients  $\pm 1$ . Finally,  $(\omega + 1)^2$  has constant coefficient 1 and  $\omega^2 - 1 = (\omega - 1)(\omega + 1)$  has constant coefficient  $-1$ ; both are reducible.  $\square$

In particular  $\omega$  has no finite irreducible factorization, not for lack of factors: it is divisible by every ordinary positive integer and by every  $\omega^\delta$  with  $0 < \delta < 1$ . Genuine infinite primes exist. The series  $1 + \sum_{k \geq 1} \omega^{1/k}$  is prime by earlier results reviewed by L’Innocente and Mantova [3, Section 1.3] (irreducibility by Berarducci, primality by Pitteloud, and primality in  $\mathbf{Oz}$  by Biljaković–Kochetov–Kuhlmann, as credited there), and  $\omega^{\sqrt{2}} + \omega + 1$  is prime by their Theorem B [3, Theorem B], which answers a question of Gonshor. Their general theory uses substantially finer invariants than the degree employed here; no result of this report depends on these primality theorems.

## 4.3 Clearing arbitrary set-sized families of denominators

**Theorem 4.6** (Global monomial clearing [01, 02, 05]). *For every set  $S \subseteq \mathbf{No}$  there is  $h > 0$  such that*

$$\omega^h \in \Pi, \quad \omega^h x \in \Pi \quad (x \in S).$$

*Consequently  $\text{Frac}(\mathbf{Oz}) = \mathbf{No}$ , where the equality means that every surreal is a quotient of two omnific integers with nonzero denominator.*

*Proof.* Take the set  $E$  obtained by uniting the supports of all elements of  $S$ . Choose  $h > 0$  strictly greater than every  $-e$  for  $e \in E$ . Multiplication by  $\omega^h$  shifts every exponent to  $h+e > 0$ ,

so all shifted normal forms belong to  $\Pi$ . Applying this to  $S = \{x\}$  gives  $x = (\omega^h x)/\omega^h$ . The reverse inclusion holds because  $\mathbf{Oz}$  is a subring of the field  $\mathbf{No}$ .  $\square$

A support unbounded below in a chosen real exponent group is still a set of surreal exponents and can be shifted above zero by a new, larger surreal exponent. The theorem does not assert that the required monomial lies in the original fixed Hahn workspace.

**Corollary 4.7** (Projective clearing and common multiples [01, 02, 05]). *Every point of  $\mathbb{P}^n(\mathbf{No})$  has homogeneous coordinates in  $\Pi^{n+1}$ ; more generally, any set-sized collection of coordinate tuples can be cleared by a single positive monomial, and a common zero of homogeneous polynomials remains a common zero after this rescaling. Every set  $S$  of nonzero omnific integers has a nonzero common multiple; in fact some nonzero  $M \in \mathbf{Oz}$  lies in  $a^k \mathbf{Oz}$  for every  $a \in S$  and ordinary  $k \geq 1$ .*

*Proof.* Apply [theorem 4.6](#) to the set of all coordinates and multiply all tuples by the resulting nonzero scalar; homogeneity preserves the equations. For the last claim apply [theorem 4.6](#) to the set  $\{a^{-k} : a \in S, k \geq 1\}$  and take  $M = \omega^h$ ; then  $M/a^k \in \Pi$ .  $\square$

This is projective *representability*, not primitivity: the coordinates of such a representative always have a common nonunit divisor, and [section 11](#) shows that some real projective directions have no primitive omnific representative at all. The common-multiple statement is an existence theorem, not an algorithm; nor is there one nonzero omnific integer divisible by every member of the proper class of nonzero omnific integers.

#### 4.4 Division with remainder is not a terminating Euclidean algorithm

**Proposition 4.8** (Ordered division [01, 02, 05]). *For  $a, b \in \mathbf{Oz}$  with  $b > 0$ , there are unique  $q, r \in \mathbf{Oz}$  such that  $a = bq + r$  and  $0 \leq r < b$ , namely  $q = \lfloor a/b \rfloor_{\mathbf{Oz}}$  and  $r = a - bq$ .*

*Proof.* The floor inequalities of [theorem 2.7](#), multiplied by  $b$ , give the remainder bounds. Conversely those bounds force  $q \leq a/b < q + 1$ , so floor uniqueness determines  $q$  and then  $r$ .  $\square$

Thus the familiar division step exists. Termination of repeated division is the missing ingredient: the positive order on  $\mathbf{Oz}$  is not well founded.

**Example 4.9** (A Euclidean algorithm that never stops [01, 02, 05]). Let  $\rho = \sqrt{2} - 1 \in (0, 1)$  and start with  $r_0 = \sqrt{2}\omega$ ,  $r_1 = \omega$ . The first quotient is 1, the remainder is  $r_2 = \rho\omega$ , and every subsequent quotient is 2:

$$r_k = \rho^{k-1}\omega \quad (k \geq 1), \quad r_{k-1} = 2r_k + r_{k+1} \quad (k \geq 2),$$

because  $1 - 2\rho = \rho^2$  and  $1/\rho = \sqrt{2} + 1 \in (2, 3)$ . Every remainder is positive and infinite, so the ordinary finite Euclidean procedure never terminates. (Source 05 starts at  $(\omega, \rho\omega)$  and sees the same tail.) A decreasing remainder is not a well-founded measure here.

**Theorem 4.10** (Pairs without a gcd [02], generalizing [01]). *Let  $0 \neq t \in \Pi$  and let  $r \in \mathbb{R} \setminus \mathbb{Q}$ . The pair  $(t, rt)$  has no greatest common divisor in  $\mathbf{Oz}$ , in the divisibility sense. In particular,  $\mathbf{Oz}$  is neither a GCD domain nor a Bézout domain.*

*Proof.* Take any common divisor  $d \in \mathbf{Oz}$ . Since  $t \neq 0$ , also  $d \neq 0$ , and there are nonzero  $u, v \in \mathbf{Oz}$  with  $t = du$  and  $rt = dv$ . Cancellation in  $\mathbf{No}$  gives  $v = ru$ . Taking constants in  $\mathcal{B}_{\mathbb{R}}$

yields  $\text{ct}(v) = r \text{ct}(u)$  with both constants ordinary integers, so irrationality of  $r$  forces both to be zero:  $u, v \in \Pi$ .

By [theorem 4.2](#) applied to  $\{u, v\}$ , choose  $\delta > 0$  with  $u/\omega^\delta, v/\omega^\delta \in \Pi$ . Then  $d' = d\omega^\delta$  is another common divisor of  $t, rt$ , and  $d \mid d'$ . But  $d' \nmid d$ , since the quotient would be  $\omega^{-\delta} \notin \mathbf{Oz}$ . Thus every common divisor can be strictly enlarged in the divisibility order, and none is a gcd. A Bézout domain would supply a gcd for each pair, namely a generator of the ideal the pair generates.  $\square$

*Remark 4.11* (A second route for the pair  $(\sqrt{2}\omega, \omega)$  [01]). Source 01 proves the instance  $t = \omega$ ,  $r = \sqrt{2}$  by degrees alone. Let  $d \neq 0$  be a common divisor,  $\theta = \deg_\omega(d)$ ; then  $0 \leq \theta \leq 1$ . If  $\theta = 1$ , both quotients are degree-zero omnific integers, hence nonzero ordinary integers, and their ratio would make  $\sqrt{2}$  rational. So  $\theta < 1$ ; with  $\eta = (\theta + 1)/2$ ,  $\omega^\eta$  is a common divisor (both quotients have exponent  $1 - \eta > 0$ ) that does not divide  $d$ , since its degree is larger. This route needs only the exponent  $(\theta + 1)/2$ , so it works in any workspace with a divisible exponent group, whereas the general [theorem 4.10](#) uses the full-class shift of [theorem 4.2](#).

[Example 4.9](#) shows at once why numerical descent is insufficient and why a division algorithm need not produce a gcd: its input has none.

## 4.5 Non-Noetherian and non-normal behavior

**Proposition 4.12** (Chains and integral closure [01, 02, 05]). *There is a strictly ascending chain of principal ideals*

$$(\omega) \subsetneq (\omega^{1/2}) \subsetneq (\omega^{1/4}) \subsetneq \cdots,$$

so  $\mathbf{Oz}$  satisfies neither the ascending chain condition on principal ideals nor the Noetherian condition. The full ring  $\mathbf{Oz}$  is not integrally closed in its fraction field: both  $\sqrt{2}$  and  $\sqrt{\omega^2 + 1}$  are integral over  $\mathbf{Oz}$ , lie in  $\text{Frac}(\mathbf{Oz})$ , and are not in  $\mathbf{Oz}$ .

*Proof.* Each generator is the square of the next, so the inclusions hold. They are strict because the reverse quotients  $\omega^{-1/2^{k+1}}$  have negative exponents. The number  $\sqrt{2} = (\sqrt{2}\omega)/\omega$  satisfies  $T^2 - 2 = 0$  and is not an integer [02, 05]. Since  $\mathbf{No}$  is real closed, let  $r = \sqrt{\omega^2 + 1} > 0$ ; it lies in  $\text{Frac}(\mathbf{Oz}) = \mathbf{No}$  and satisfies  $T^2 - (\omega^2 + 1) = 0$ . If  $r \in \mathbf{Oz}$ , then  $(r - \omega)(r + \omega) = 1$  makes both factors units, hence elements of  $\{1, -1\}$  with equal signs; their difference  $2\omega$  would be 0, a contradiction [01].  $\square$

In particular the full omnific ring is not a UFD or a PID. Its factorization theory cannot be replaced by an ordinary Euclidean algorithm or by an unstated gcd assumption. The modern theory in [3] is substantially subtler. Conway's refinement question asks whether  $ab = cd$  in  $\mathbf{Oz}$  always admits  $a = ef$ ,  $b = gh$ ,  $c = eg$ ,  $d = fh$  with  $e, f, g, h \in \mathbf{Oz}$ ; this is a refinement property, not atomicity or the existence of gcds, and none of the preceding arguments refutes it. [Appendix D](#) records why no current open-status claim about it is made.

## 5 Equational transfer and linear Diophantine equations

### 5.1 Existence is exactly ordinary existence

For a finite family of polynomials  $F_1, \dots, F_r$  over a ring  $R$ , write  $V_R(F_1, \dots, F_r)$  for its simultaneous zero set in  $R^n$ . All exponents of the variables are ordinary nonnegative integers and only finitely many monomials occur; equations with variable exponents, surreal exponentiation, or class-sized sums are not polynomial systems in this sense.

**Theorem 5.1** (Retraction theorem for equations [01, 02, 05]). *Let  $F_1, \dots, F_r \in \mathbb{Z}[X_1, \dots, X_n]$ . Coordinatewise constant coefficient gives a retraction, with the inclusion as section,*

$$V_{\mathbf{Oz}}(F_1, \dots, F_r) \longrightarrow V_{\mathbb{Z}}(F_1, \dots, F_r).$$

*In particular,*

$$V_{\mathbf{Oz}}(F_1, \dots, F_r) \neq \emptyset \iff V_{\mathbb{Z}}(F_1, \dots, F_r) \neq \emptyset. \quad (5.1)$$

*The same equivalence holds for positive existential formulas in the ring language, evaluated at ordinary integer parameters.*

*Proof.* For every polynomial  $F$  over  $\mathbb{Z}$  and  $\mathbf{x} \in \mathbf{Oz}^n$ ,  $\text{ct}(F(\mathbf{x})) = F(\text{ct}(x_1), \dots, \text{ct}(x_n))$ , because  $\text{ct}$  is a ring homomorphism fixing all coefficients. Thus the constant tuple of an omnific solution is an integer solution. Conversely, an integer solution is already an omnific solution, and applying  $\text{ct}$  to it does nothing.

A positive existential formula is built from equalities using finite conjunction, finite disjunction, and existential quantification. Ring homomorphisms preserve such formulas, by induction on their formation, with witnesses mapped coordinatewise. Apply this to  $\text{ct}$  and to the inclusion of  $\mathbb{Z}$  in  $\mathbf{Oz}$ .  $\square$

The theorem answers a question about existence, not about cardinalities or complete solution sets. For  $\mathbf{a} \in V_{\mathbb{Z}}$ , its *constant-coefficient fiber* is

$$V_{\mathbf{a}}(\mathbf{Oz}) = \{\mathbf{a} + \mathbf{u} : \mathbf{u} \in \Pi^n, F_j(\mathbf{a} + \mathbf{u}) = 0 \text{ for all } j\}. \quad (5.2)$$

For  $Y = X^2$  the fiber above  $(0, 0)$  already contains  $(t, t^2)$  for every  $t \in \Pi$ . These are not infinitesimal deformations: every nonzero coordinate of  $\mathbf{u}$  is infinite. We call a system *rigid* when every fiber is its ordinary base point, that is, when  $V_{\mathbf{Oz}} = V_{\mathbb{Z}}$  [02].

**Corollary 5.2** (Hilbert’s tenth problem [01, 02, 05]). *There is no algorithm deciding, from a finite list of ordinary integer polynomials, whether it has a common zero in  $\mathbf{Oz}$ . The yes-instances are exactly the ordinary-integer yes-instances of Hilbert’s tenth problem, and form a computably enumerable set that is complete for computably enumerable sets under many-one reductions.*

*Proof.* Equation (5.1) identifies the two decision problems on exactly the same finite inputs. The undecidability and completeness of the ordinary problem are the Davis–Putnam–Robinson–Matiyasevich (MRDP) theorem and its standard consequences [7, 8, 9, 10]; one passes between the natural-number and integer formulations by the four-square representation of nonnegative integers, and conjunctions are combined as sums of squares. Enumerating ordinary integer tuples semidecides the omnific yes-instances; no enumeration of the proper class  $\mathbf{Oz}$  is involved.  $\square$

This corollary concerns *ordinary integer coefficients*. A decision problem allowing arbitrary omnific coefficients requires an explicit finite representation scheme and is a different question (section 12).

## 5.2 Why signs and nonvanishing do not transfer

Consider  $(X + 1)^2 - 2(Y + 1)^2 = 0$ . It has no solution in  $\mathbb{N}^2$ , since  $X + 1$  and  $Y + 1$  would be positive integers with ratio  $\sqrt{2}$ . However,  $X = \sqrt{2}\omega - 1$ ,  $Y = \omega - 1$  is a solution with both coordinates positive in  $\mathbf{Oz}$ . Its constant tuple is  $(-1, -1)$ , an ordinary solution of the equation without positivity constraints.

**Proposition 5.3** (Retraction closure [01, 02, 05]). *If  $D \subseteq \mathbf{Oz}^n$  is defined by a positive existential ring formula with ordinary integer parameters, then  $\mathbf{x} \in D$  implies  $\text{ct}(\mathbf{x}) \in D$ . Consequently none of*

$$\mathbf{Oz} \setminus \{0\}, \quad \mathbf{Oz}^{>0}, \quad \mathbf{Oz}^{\geq 0}$$

*is so definable.*

*Proof.* Apply  $\text{ct}$  to the free variables and to all existential witnesses, as in [theorem 5.1](#). But  $\omega$  is nonzero and positive while  $\text{ct}(\omega) = 0$ ; and  $\omega - 1 > 0$  whereas  $\text{ct}(\omega - 1) = -1 < 0$ .  $\square$

Thus “nonzero” cannot be harmlessly added to an equational transfer statement, and the retraction is not an elementary embedding of ordered rings. Encoding nonvanishing by  $XY = 1$  would restrict  $X$  to  $\{1, -1\}$ , not to all nonzero omnific integers. The qualification on coefficients matters as well: a predicate with a fixed infinite parameter need not be preserved by a map that changes that parameter. For coefficients outside  $\mathbb{Z}$ , specialization gives a necessary residual system whose solvability need not lift ([section 12](#)).

### 5.3 A complete solution criterion for linear systems

**Theorem 5.4** (Smith normal form over the omnific integers [01, 02, 05]). *Let  $M \in \mathbb{Z}^{m \times n}$  and  $\mathbf{b} \in \mathbf{Oz}^m$ . Choose unimodular integer matrices  $U, V$  with*

$$UMV = \text{diag}(d_1, \dots, d_r, 0), \quad d_i > 0,$$

*in rectangular Smith normal form, and put  $\mathbf{c} = U\mathbf{b}$ . Then  $M\mathbf{x} = \mathbf{b}$  is solvable in  $\mathbf{Oz}^n$  if and only if*

$$d_i \mid \text{ct}(c_i) \quad (1 \leq i \leq r), \quad c_i = 0 \quad (r < i \leq m). \quad (5.3)$$

*All solutions are  $\mathbf{x} = V\mathbf{y}$ , where  $y_i = c_i/d_i$  for  $1 \leq i \leq r$  and  $y_{r+1}, \dots, y_n \in \mathbf{Oz}$  are arbitrary.*

*Proof.* Unimodular integer matrices and their inverses act on omnific tuples. The substitution  $\mathbf{x} = V\mathbf{y}$  transforms the system into  $\text{diag}(d_1, \dots, d_r, 0)\mathbf{y} = \mathbf{c}$ . For each nonzero diagonal entry, division is possible precisely under [theorem 3.1](#). Zero rows require the full equation  $c_i = 0$ , not merely its constant coefficient. The remaining coordinates are unconstrained.  $\square$

**Theorem 5.5** (Ordinary right-hand sides [05]). *Let  $M \in \mathbb{Z}^{m \times n}$  and  $\mathbf{b} \in \mathbb{Z}^m$ . All solutions in  $\mathbf{Oz}^n$  of  $M\mathbf{x} = \mathbf{b}$  are exactly*

$$\mathbf{x} = \mathbf{a} + \mathbf{u}, \quad \mathbf{a} \in \mathbb{Z}^n, \quad M\mathbf{a} = \mathbf{b}, \quad \mathbf{u} \in \Pi^n, \quad M\mathbf{u} = 0.$$

*If  $\mathbf{v}_1, \dots, \mathbf{v}_e \in \mathbb{Q}^n$  is a basis of  $\ker(M : \mathbb{Q}^n \rightarrow \mathbb{Q}^m)$ , the last condition is equivalent to  $\mathbf{u} = \sum_j h_j \mathbf{v}_j$  with uniquely determined  $h_j \in \Pi$ . In particular, if  $M$  has full column rank, all solutions are ordinary.*

*Proof.* Separate each coordinate into its integer constant and its member of  $\Pi$ . The integer and positive-support components of  $M\mathbf{x} - \mathbf{b}$  vanish separately. Gaussian elimination over  $\mathbb{Q}$  works with entries in the  $\mathbb{Q}$ -vector space  $\Pi$ , since nonzero rationals act invertibly on it; applying the same row operations and free-variable parametrization as over  $\mathbb{Q}$  gives the basis description and its uniqueness.  $\square$

For example, every solution of  $2x + 3y = 1$  is  $x = -1 + 3t$ ,  $y = 1 - 2t$  with  $t \in \mathbf{Oz}$ , and every solution of  $6x + 10y = 14$  is  $x = 4 + 5t$ ,  $y = -1 - 3t$  with  $t \in \mathbf{Oz}$ . The same formulas as over  $\mathbb{Z}$  work, but the parameter class is larger. The equation  $2x = \omega + 1$  has no solution, whereas  $2x = \omega + 2$  has the unique solution  $x = \omega/2 + 1$ .

## 6 Rigidity of constant products: binary forms, Pell, norms

### 6.1 The constant-product principle

**Lemma 6.1** (Constant products [01, 02, 05]). *If  $b_1, \dots, b_s \in \mathcal{B}_{\mathbb{C}}$  and  $b_1 \cdots b_s = c \in \mathbb{C}^{\times}$ , then every  $b_i$  belongs to  $\mathbb{C}^{\times}$ .*

*Proof.* Each  $b_i$  has inverse  $c^{-1} \prod_{j \neq i} b_j$  in  $\mathcal{B}_{\mathbb{C}}$ ; apply [proposition 2.5](#). Equivalently, their nonnegative degrees have sum zero, so every degree is zero.  $\square$

The importance of  $\mathcal{B}_{\mathbb{C}}$  is that linear factors may have irrational or complex coefficients. They need not themselves be omnific integers, but they have no negative exponents, so the unit argument applies.

**Theorem 6.2** (Exact fibers of decomposable equations [02]; case (b) [01]). *Let*

$$F(\mathbf{X}) = \prod_{j=1}^s L_j(\mathbf{X})^{e_j}, \quad L_j(\mathbf{X}) = \beta_j + \ell_j(\mathbf{X}),$$

where  $\beta_j \in \mathbb{C}$ , each  $\ell_j$  is a complex linear form in  $n$  variables, and each  $e_j \geq 1$  is an ordinary integer (a constant factor can be absorbed into one  $L_j$ ). Fix  $c \in \mathbb{C}^{\times}$  and put

$$\mathcal{K} = \{\mathbf{v} \in \mathbb{R}^n : \ell_j(\mathbf{v}) = 0 \text{ for every } j\}.$$

(a) *The solutions of  $F(\mathbf{x}) = c$  in  $\mathbf{Oz}^n$  are exactly*

$$\mathbf{x} = \mathbf{a} + \mathbf{u}, \quad \mathbf{a} \in \mathbb{Z}^n, \quad F(\mathbf{a}) = c, \quad \mathbf{u} \in \Pi^n, \quad \ell_j(\mathbf{u}) = 0 \text{ for all } j. \quad (6.1)$$

*Equivalently, at each exponent the real coefficient vector of  $\mathbf{u}$  lies in  $\mathcal{K}$ ; if  $\mathbf{v}_1, \dots, \mathbf{v}_e$  is a real basis of  $\mathcal{K}$ , the fiber above  $\mathbf{a}$  is  $\mathbf{a} + \{\sum_k t_k \mathbf{v}_k : t_k \in \Pi\}$ . If  $\mathcal{K} = 0$ , all omnific solutions are ordinary.*

(b) *If the linear parts  $\ell_1, \dots, \ell_s$  have column rank  $n$  over  $\mathbb{C}$ , then every solution  $\mathbf{x} \in \mathcal{B}_{\mathbb{C}}^n$  of  $F(\mathbf{x}) = c$  lies in  $\mathbb{C}^n$ .*

*Proof.* For  $\mathbf{x} \in \mathcal{B}_{\mathbb{C}}^n$ , every  $L_j(\mathbf{x})$  is in  $\mathcal{B}_{\mathbb{C}}$ , and if  $F(\mathbf{x}) = c \neq 0$ , [lemma 6.1](#) makes each  $L_j(\mathbf{x})$  constant.

(a) Write  $\mathbf{x} = \mathbf{a} + \mathbf{u}$  coordinatewise with  $\mathbf{a} \in \mathbb{Z}^n$  and  $\mathbf{u} \in \Pi^n$ . Then  $L_j(\mathbf{x}) = L_j(\mathbf{a}) + \ell_j(\mathbf{u})$ , where the first summand is constant and the second lies in  $\Pi_{\mathbb{C}}$ ; uniqueness of normal forms forces  $\ell_j(\mathbf{u}) = 0$ , and specialization gives  $F(\mathbf{a}) = c$ . Conversely those conditions give  $L_j(\mathbf{a} + \mathbf{u}) = L_j(\mathbf{a})$  and hence  $F(\mathbf{a} + \mathbf{u}) = c$ . At each exponent of the (set-sized, reverse-well-ordered) union of the coordinate supports, the equations  $\ell_j(\mathbf{u}) = 0$  say exactly that the real coefficient vector  $\mathbf{u}_{\gamma} \in \mathbb{R}^n$  at that exponent satisfies  $\ell_j(\mathbf{u}_{\gamma}) = 0$  for all  $j$ , that is, lies in  $\mathcal{K}$ . Expressing each coefficient vector in the basis  $\mathbf{v}_k$  produces  $t_k$  whose supports lie in the same union; if  $\mathcal{K} = 0$ , every coefficient vector vanishes.

(b) Select  $n$  of the  $\ell_j$  whose coefficient matrix is invertible over  $\mathbb{C}$ , subtract the constant terms and solve this ordinary linear system: all coordinates are complex constants.  $\square$

The kernel hypothesis in (a) is exactly what matters. A single-factor form can have infinite fibers,

$$(x - y)^2 = 1 \iff x = y \pm 1, \quad y \in \mathbf{Oz},$$

and the two-variable equation  $X^d = 1$  forces  $X$  to be constant but says nothing about  $Y$ . On the zero level, even independent factors permit infinite points, since the unit argument cannot be applied to a product equal to zero:  $(\sqrt{2}\omega)^2 - 2\omega^2 = 0$ .

## 6.2 Binary forms, Pell equations, and conics

**Theorem 6.3** (Binary-form rigidity [05], generalizing [01, 02]). *Let  $F \in \mathbb{C}[X, Y]$  be homogeneous with at least two distinct projective linear factors over  $\mathbb{C}$ , and let  $c \in \mathbb{C}^\times$ . If  $x, y \in \mathcal{B}_{\mathbb{C}}$  satisfy  $F(x, y) = c$ , then  $x, y \in \mathbb{C}$ . In particular, for  $F \in \mathbb{Z}[X, Y]$  and  $c \in \mathbb{Z} \setminus \{0\}$ ,*

$$\{(x, y) \in \mathbf{Oz}^2 : F(x, y) = c\} = \{(x, y) \in \mathbb{Z}^2 : F(x, y) = c\}.$$

*Proof.* Factor  $F = \alpha \prod_{j=1}^d L_j$  over  $\mathbb{C}$  with  $\alpha \in \mathbb{C}^\times$  and nonzero linear forms repeated according to multiplicity. Two distinct projective factors have linearly independent coefficient vectors, so the linear parts have rank two; apply [theorem 6.2\(b\)](#). Constant omnific coordinates are integers.  $\square$

Sources 01 and 02 proved the statement for  $F \in \mathbb{Z}[X, Y]$ ,  $c \in \mathbb{Z}$  and  $\mathbf{Oz}$ -points; the complex form is 05's. The hypothesis excludes a pure power  $\alpha L^d$ , and that exclusion is necessary: fixing  $L(x, y)$  to a constant leaves a free infinite direction in the kernel of  $L$ . The requirement  $c \neq 0$  is also necessary, since zero levels contain lines through the origin. The hypothesis is weaker than squarefreeness in degree at least three:  $X^2Y$  qualifies.

**Corollary 6.4** (Thue levels and equal powers [01, 02, 05]). *If an integer binary form has at least two distinct projective factors, its nonzero ordinary constant levels have no additional omnific points. This applies to every irreducible binary form over  $\mathbb{Q}$  of degree at least two, hence to Thue equations, and to  $x^d - y^d = c$  with  $d \geq 2$  and  $c \in \mathbb{Z} \setminus \{0\}$ .*

*Proof.* In characteristic zero an irreducible binary form of degree at least two has distinct projective roots over  $\mathbb{C}$ , and  $X^d - Y^d$  has distinct linear factors. Apply [theorem 6.3](#).  $\square$

The conclusion identifies the omnific solution class with the classical integer solution set; it is not a new proof of ordinary finiteness, gives no bound for the ordinary solutions, and supplies no algorithm for them. No finiteness theorem of Thue or Baker is used.

**Corollary 6.5** (Pell rigidity [01, 02, 05]). *For ordinary integers  $D \neq 0$  and  $c \neq 0$ , every solution in  $\mathbf{Oz}^2$  of  $x^2 - Dy^2 = c$  is an ordinary integer solution. In particular*

$$\{(u, v) \in \mathbf{Oz}^2 : u^2 - 2v^2 = 1\} = \{(u, v) \in \mathbb{Z}^2 : u^2 - 2v^2 = 1\}.$$

*Proof.* Over  $\mathbb{C}$ , the factors  $X - \sqrt{D}Y$  and  $X + \sqrt{D}Y$  are distinct because  $D \neq 0$ . Apply [theorem 6.3](#).  $\square$

Source 02 states this corollary for  $D > 0$  only; its proof, like those of 01 and 05, uses only  $D \neq 0$ , and the conclusion holds whether or not  $D$  is a square. “Rigid” means no new points, not a finite number of old points: the classical Pell equation has ordinary solutions of arbitrarily large finite size, but no single infinite omnific solution, and there is no “Pell solution at index  $\omega$ ” obtained by substituting an omnific index into a finite recurrence. This is a first warning against treating  $\mathbf{Oz}$  as an elementary extension of ordinary integer arithmetic.

**Example 6.6** (The full class for  $D = 2$ ,  $c = 1$  [05]). Let  $X_k + Y_k\sqrt{2} = (3 + 2\sqrt{2})^k$  for ordinary  $k \geq 0$ . Every omnific solution of  $x^2 - 2y^2 = 1$  is  $(\pm X_k, \pm Y_k)$  with independent signs. Indeed, [corollary 6.5](#) first makes  $x, y$  ordinary. After changing signs suppose  $x, y > 0$ ; then  $y \geq 2$ , since  $y = 1$  would need  $x^2 = 3$ . Multiplication by  $3 - 2\sqrt{2}$  gives the integer Pell pair  $X = 3x - 4y$ ,  $Y = 3y - 2x$  with  $X > 0$  (as  $x > \sqrt{2}y > 4y/3$ ) and  $0 \leq Y < y$  (as  $x^2 = 2y^2 + 1 \leq 9y^2/4$  and  $x > y$ ). Ordinary descent on  $y$  reaches  $(1, 0)$ , and reversing it gives the asserted powers.

In the full surreal field the contrast is visible without solving a recurrence [01]. For  $D > 0$ ,  $c \neq 0$  and  $s \in \mathbf{No}^\times$ , the pair  $x = (s + c/s)/2$ ,  $y = (s - c/s)/(2\sqrt{D})$  satisfies  $x^2 - Dy^2 = c$ . Taking  $s = \omega$  produces negative-exponent terms, so neither coordinate is an omnific integer, and discarding those terms changes the norm from  $c$  to zero. A surreal solution cannot be turned into an omnific solution by unverified truncation.

**Proposition 6.7** (The zero Pell fiber [01]). *If  $D$  is a positive nonsquare ordinary integer, the solutions of  $x^2 - Dy^2 = 0$  in  $\mathbf{Oz}^2$  are exactly  $(\pm\sqrt{D}t, t)$  with  $t \in \Pi$ .*

*Proof.* Factor in  $\mathbf{No}$  to obtain  $x = \pm\sqrt{D}y$ . Write  $y = m + j$  with  $m \in \mathbb{Z}$  and  $j \in \Pi$ . The constant coefficient of  $x$  is  $\pm\sqrt{D}m$ , which is an integer only if  $m = 0$ . Conversely,  $\Pi$  is an  $\mathbb{R}$ -vector space, so all displayed pairs belong to  $\mathbf{Oz}^2$  and solve the equation.  $\square$

**Corollary 6.8** (Nonsingular central conics [05]). *Let  $F(X, Y) = q(X, Y) + \ell(X, Y) + d \in \mathbb{R}[X, Y]$ , where the quadratic form  $q$  has rank two, and let  $\mathbf{h} \in \mathbb{R}^2$  be the center, so that  $F(\mathbf{h} + \mathbf{u}) = q(\mathbf{u}) + F(\mathbf{h})$ . If  $F(\mathbf{h}) \neq 0$ , every solution of  $F = 0$  in  $\mathcal{B}_{\mathbb{R}}^2$  is constant; for integer coefficients, every omnific solution is ordinary.*

*Proof.* A rank-two binary quadratic form has two distinct projective factors over  $\mathbb{C}$ . The translated coordinates  $\mathbf{u} = (x, y) - \mathbf{h}$  lie in  $\mathcal{B}_{\mathbb{R}}^2$  and satisfy  $q(\mathbf{u}) = -F(\mathbf{h}) \neq 0$ . Apply [theorem 6.3](#) and translate back.  $\square$

This covers ellipses and hyperbolas with a nonzero centered level. Parabolas behave differently:  $Y = X^2$  has the infinite family  $(t, t^2)$ .

*Remark 6.9* (Power differences with a common factor [01]). Source 01 proved that every omnific solution of  $x^m - y^n = c \neq 0$  is ordinary when  $d = \gcd(m, n) > 1$ : the binary form  $X^d - Y^d$  in  $X = x^{m/d}$ ,  $Y = y^{n/d}$  is rigid by [theorem 6.3](#), and an element of  $\mathbf{Oz}$  with an ordinary constant positive power is ordinary (by degree additivity if the power is nonzero, by the domain property otherwise). Source 01 stated explicitly that this argument does not classify coprime cases such as  $x^2 - y^3 = 1$ . All exponent pairs are covered by [theorem 7.4](#) of source 05.

### 6.3 Number-field norm equations

**Theorem 6.10** (Norm-form rigidity [01, 02]). *Let  $K/\mathbb{Q}$  be a number field of degree  $d$ , let  $e_1, \dots, e_d$  be a  $\mathbb{Q}$ -basis, and let  $N(\mathbf{X}) = N_{K/\mathbb{Q}}(\sum_{j=1}^d e_j X_j)$ . For every  $c \in \mathbb{Q}^\times$ , every solution  $\mathbf{x} \in \mathbf{Oz}^d$  of  $N(\mathbf{x}) = c$  has all coordinates in  $\mathbb{Z}$ . If the basis consists of algebraic integers (for example a power basis of an algebraic integer  $\theta$  generating  $K$ , or an integral basis), then  $N \in \mathbb{Z}[\mathbf{X}]$ , and the statement applies to the usual integer norm equations.*

*Proof.* The  $d$  embeddings  $\sigma_1, \dots, \sigma_d : K \rightarrow \mathbb{C}$  give

$$N(\mathbf{X}) = \prod_{i=1}^d \left( \sum_{j=1}^d \sigma_i(e_j) X_j \right).$$

The embedding matrix  $(\sigma_i(e_j))$  is invertible: for a power basis  $1, \theta, \dots, \theta^{d-1}$  of a primitive element of the separable extension  $K/\mathbb{Q}$  it is the Vandermonde matrix of the distinct conjugates of  $\theta$ , and a basis change over  $\mathbb{Q}$  preserves invertibility. The forms therefore have zero common kernel and [theorem 6.2](#) applies. If the  $e_j$  are algebraic integers, each coefficient of  $N$  is fixed by the Galois group of a normal closure, hence rational, and is an integer polynomial in conjugates of the  $e_j$ , hence an algebraic integer; so it is an ordinary integer [02].  $\square$

For example, consider the norm form for  $K = \mathbb{Q}(\sqrt[3]{2})$ :

$$N(X + Y\sqrt[3]{2} + Z\sqrt[3]{4}) = X^3 + 2Y^3 + 4Z^3 - 6XYZ.$$

For every  $c \in \mathbb{Z} \setminus \{0\}$ , every omnific solution of  $X^3 + 2Y^3 + 4Z^3 - 6XYZ = c$  is an ordinary integer triple. This is a three-variable rigidity statement on an unbounded real surface, unlike the indefinite quadratic forms of [section 8](#): dimension alone does not force infinite omnific points. The identity follows from the product over the three conjugates or from the determinant of multiplication by  $X + Y\theta + Z\theta^2$ ; the suites of 01 and 02 check it exactly.

## 6.4 An algebraic-geometric formulation

Let  $V$  be an affine complex variety whose coordinate ring  $\mathbb{C}[V]$  is generated, as a  $\mathbb{C}$ -algebra, by units. Every  $\mathcal{B}_{\mathbb{C}}$ -point of  $V$  is then a constant  $\mathbb{C}$ -point [01, 02]. Indeed, a  $\mathbb{C}$ -algebra map  $\mathbb{C}[V] \rightarrow \mathcal{B}_{\mathbb{C}}$  sends each generating unit to an element of  $\mathcal{B}_{\mathbb{C}}^{\times} = \mathbb{C}^{\times}$ , so its entire image lies in  $\mathbb{C}$ . This includes closed subvarieties of an algebraic torus  $(\mathbb{C}^{\times})^r$  in their very-affine coordinate presentation. The norm argument is a concrete linear-coordinate instance: a nonzero product relation forces its factors into a constant torus. It is not a claim that every affine variety is constant over  $\mathcal{B}_{\mathbb{C}}$ ; the affine line has the full class  $\mathcal{B}_{\mathbb{C}}$  of points.

## 7 Euler derivations: separated powers and unimodular Fermat

Binary forms can be factored into linear forms, but equations such as  $Y^2 = X^3 + 17$  cannot be handled that way. The following local tool, due to source 05, gives short arguments without any finite-support hypothesis.

### 7.1 Support-preserving Euler derivations

Fix a set-sized ordered divisible group  $H \subseteq \mathbf{No}$  and a  $\mathbb{Q}$ -linear map  $\lambda : H \rightarrow \mathbb{R}$ . On the real or complex Hahn field with exponents in  $H$ , define

$$\partial_{\lambda} \left( \sum_{h \in H} a_h \omega^h \right) = \sum_{h \in H} \lambda(h) a_h \omega^h. \quad (7.1)$$

This is a local, freely chosen Euler derivation. It is not a canonical surreal derivation and not the Berarducci–Mantova derivation.

**Lemma 7.1** (Euler derivations [05]). *The map  $\partial_{\lambda}$  is a derivation, kills every real or complex constant, and never introduces a new supported exponent; so  $\deg_{\omega}(\partial_{\lambda} f) \leq \deg_{\omega}(f)$  whenever  $\partial_{\lambda} f \neq 0$ . It restricts to derivations of  $\mathcal{B}_{\mathbb{R}, H}$  and  $\mathcal{B}_{\mathbb{C}, H}$ .*

*Proof.* The new support is a subset of the old one and is thus reverse well ordered. Additivity is immediate, and  $\lambda(0) = 0$  kills constants. For a fixed exponent  $h$ , the coefficient convolution is finite and  $\lambda(a + b) = \lambda(a) + \lambda(b)$ , so

$$[\omega^h] \partial_{\lambda}(fg) = \sum_{a+b=h} (\lambda(a) + \lambda(b)) f_a g_b = [\omega^h] ((\partial_{\lambda} f)g + f(\partial_{\lambda} g)).$$

This proves the Leibniz rule coefficientwise; no interchange of conditionally convergent sums is involved. The degree and restriction statements follow from support containment.  $\square$

**Lemma 7.2** (Detection of nonconstants [05]). *Every nonconstant element of a Hahn field with exponent group  $H$  is detected by some  $\partial_\lambda$ . If  $f \in \mathcal{B}_{\mathbb{C},H}$  has positive leading exponent  $\alpha$ , one may choose  $\lambda(\alpha) = 1$ ; then  $\deg_\omega(\partial_\lambda f) = \alpha$ .*

*Proof.* Choose a nonzero supported exponent  $h$ . A  $\mathbb{Q}$ -linear functional on  $\mathbb{Q}h$  taking  $h$  to 1 extends to  $H$  by extending a rational basis;  $H$  is a set, so choice for sets suffices. The derivative has nonzero coefficient at  $h$ . Choosing the leading exponent gives the final assertion.  $\square$

For  $\partial = \partial_\lambda$  define the two-term Wronskian  $W_\partial(f, g) = f \partial g - g \partial f$ . For  $f \neq 0$  it vanishes exactly when  $\partial(g/f) = 0$ ; it need not detect whether  $g/f$  is an absolute constant for one fixed  $\partial$ , which is why [lemma 7.2](#) lets the derivation be chosen for the given ratio.

**Lemma 7.3** (A strict Wronskian bound [05]). *For nonzero  $f, g \in \mathcal{B}_{\mathbb{C},H}$  with  $W_\partial(f, g) \neq 0$ ,  $\deg_\omega W_\partial(f, g) \leq \deg_\omega f + \deg_\omega g$ . If  $\deg_\omega f = \deg_\omega g = \alpha$ , the inequality is strict:  $\deg_\omega W_\partial(f, g) < 2\alpha$ .*

*Proof.* The weak bound follows from [lemmas 2.4](#) and [7.1](#). If the leading coefficients of  $f, g$  are  $a, b$  at exponent  $\alpha$ , the two possible contributions at  $2\alpha$  are  $a\lambda(\alpha)b$  and  $b\lambda(\alpha)a$ , which cancel; every other contribution has smaller exponent. A nonzero Wronskian has a greatest supported exponent, by reverse well-ordering, so its degree is strictly below  $2\alpha$ ; one does not replace that maximum by an unattained supremum.  $\square$

## 7.2 Separated powers

**Theorem 7.4** (Separated-power rigidity [05]). *Let  $a, b, c \in \mathbb{R} \setminus \{0\}$  and let  $m, n \geq 2$  be ordinary integers. Every solution in  $\mathcal{B}_{\mathbb{R}}^2$  of*

$$ax^m + by^n = c \tag{7.2}$$

*has  $x, y \in \mathbb{R}$ . Consequently, if  $a, b, c$  are nonzero ordinary integers, every solution in  $\mathbf{Oz}^2$  is an ordinary integer solution.*

*Proof.* If one coordinate is constant, the other is constant by [proposition 2.8](#). Suppose both are nonconstant, with  $\alpha = \deg_\omega x > 0$  and  $\beta = \deg_\omega y > 0$ . Cancellation of the positive leading degree in (7.2) forces  $m\alpha = n\beta$ . Take a common set-sized workspace  $H$  for  $x, y$  ([lemma 2.6](#)) and choose  $\partial = \partial_\lambda$  with  $\lambda(\alpha) = 1$ , so  $\partial x \neq 0$ . Differentiating gives

$$amx^{m-1}\partial x + bny^{n-1}\partial y = 0, \tag{7.3}$$

and since  $x \neq 0$  also  $\partial y \neq 0$ .

We work in  $\mathcal{B}_{\mathbb{R},H}$ , in which  $a, b, c, m, n$  are units. Modulo the principal ideal  $(x^{m-1})$ , (7.2) says  $by^n = c$ , so the image of  $y$  is a unit; reducing (7.3) then shows that  $\partial y$  vanishes in the quotient. Equivalently  $\partial y = x^{m-1}q$  with  $q \in \mathcal{B}_{\mathbb{R},H}$ , and  $q \neq 0$  because  $\partial y \neq 0$ . Nonnegative degree and [lemma 7.1](#) give

$$(m-1)\alpha \leq \deg_\omega(\partial y) \leq \beta = \frac{m}{n}\alpha.$$

Since  $\alpha > 0$ , this gives  $mn - m - n \leq 0$ , that is  $(m-1)(n-1) \leq 1$ , which for  $m, n \geq 2$  forces  $m = n = 2$ . In that case  $aX^2 + bY^2$  has two distinct projective linear factors over  $\mathbb{C}$  because  $ab \neq 0$ , and [theorem 6.3](#) contradicts the assumed nonconstancy. Constant omnific coordinates are integers.  $\square$

The quotient-ring step replaces any appeal to unique factorization: if  $y$  becomes a unit modulo  $x^{m-1}$ , a product of  $\partial y$  with a unit can vanish there only if  $\partial y$  itself does. The argument is valid in a non-Noetherian ring and for supports of arbitrary set-sized order type.

*Remark 7.5* (A direct divisibility identity [merge]). The divisibility  $x^{m-1} \mid \partial y$  also follows from an explicit identity. Put  $f = ax^m + by^n$ . For every derivation  $\partial$ ,

$$ax^{m-1}(nx\partial y - my\partial x) = n f \partial y - y \partial f,$$

as one sees by expanding  $\partial f = amx^{m-1}\partial x + bny^{n-1}\partial y$ : the right side is  $anx^m\partial y + bny^n\partial y - amx^{m-1}y\partial x - bny^n\partial y$ . On a solution of (7.2),  $f = c$  and  $\partial f = 0$ , so  $nc\partial y = x^{m-1} \cdot a(nx\partial y - my\partial x)$ , and since  $nc$  is a unit of  $\mathcal{B}_{\mathbb{R},H}$ ,  $x^{m-1}$  divides  $\partial y$ . This route avoids the quotient ring. The suite of 05 checks a related identity only for  $a = 1, b = -1$ ; the identity above was checked symbolically for general  $a, b$  and  $2 \leq m, n \leq 7$  when this report was assembled.

**Corollary 7.6** (Mordell and Fermat curves [05]). *The following equations have no infinite omnific solutions; their omnific solution classes equal their ordinary integer solution sets:*

$$\begin{aligned} x^m - y^n &= c \quad (m, n \geq 2, c \in \mathbb{Z} \setminus \{0\}), \\ y^2 &= x^3 + k \quad (k \in \mathbb{Z} \setminus \{0\}), \quad x^n + y^n = 1 \quad (n \geq 2). \end{aligned}$$

*Proof.* Use  $(a, b) = (1, -1)$  or  $(-1, 1)$  as appropriate in theorem 7.4, and  $(a, b, c) = (1, 1, 1)$  for the last equation.  $\square$

In particular  $Y^2 = X^3 + 1$ , which source 01 listed as an open question (section 14.1), has only its ordinary integer points in  $\mathbf{Oz}$ . The theorem reduces the omnific problem to ordinary integers; it does not by itself bound or list the integer solutions. A finite computation can verify particular solutions of  $y^2 = x^3 + 1$ , such as  $(-1, 0), (0, \pm 1), (2, \pm 3)$ , but a finite search is not a proof that such a list is complete, and the suite of 05 checks only the displayed identities.

The condition  $c \neq 0$  cannot be removed:  $x = t^n, y = t^m$  solves  $x^m - y^n = 0$  for every  $t \in \mathbf{Oz}$ . Nor can an exponent equal one:  $y = x^m - c$  is an infinite family. The theorem treats fixed ordinary exponents and says nothing about variable omnific exponents. A general elliptic equation  $y^2 = x^3 + ax + b$  with  $a \neq 0$  is outside it: the differentiated equation has additional terms, and the divisibility by a high power of  $x$  no longer follows. This is a boundary of the proof, not a reason to extend the statement to all elliptic curves.

### 7.3 Unimodular Fermat rigidity

The homogeneous Fermat equation has abundant infinite solutions (example 11.3), none of them primitive. An explicit unit-ideal condition excludes nonconstant solutions for every exponent at least three. We use the following terms, which sources 01 and 05 define identically in substance.

**Definition 7.7** (Primitive and unimodular [01, 05]). A nonzero tuple  $(x_1, \dots, x_r)$  over a commutative ring  $R$  is *unimodular* if  $\sum_i u_i x_i = 1$  for some  $u_i \in R$ , that is,  $(x_1, \dots, x_r) = R$ . Over the domain  $\mathbf{Oz}$  it is *primitive* if its entries have no common nonunit divisor.

Unimodularity implies primitivity. The converse is not assumed in the non-PID ring  $\mathbf{Oz}$ , and no Bézout hypothesis is used; whenever a tuple is claimed unimodular, an explicit identity is given.

**Lemma 7.8** (Pairwise comaximality [05]). *Let  $R$  be a commutative ring and suppose  $x^n + y^n = z^n$  with  $n \geq 1$  and  $(x, y, z) = R$ . Then  $(x, y) = (y, z) = (z, x) = R$ , and powers of any of these pairs are again comaximal.*

*Proof.* In  $R/(x, y)$  the element  $z$  is nilpotent, as  $z^n = 0$ , and a unit, because a unit-ideal witness reduces to  $wz = 1$ . The quotient is therefore zero. The other pairs are similar. If  $au + bv = 1$ , the binomial expansion of  $(au + bv)^{r+s-1}$  lies in  $(u^r, v^s)$ .  $\square$

**Theorem 7.9** (Unimodular Fermat rigidity [05]). *Let  $n \geq 3$  be an ordinary integer. Suppose  $x, y, z \in \mathcal{B}_{\mathbb{C}}$  are nonzero,*

$$x^n + y^n = z^n, \quad (x, y, z) = \mathcal{B}_{\mathbb{C}}.$$

*Then  $x, y, z \in \mathbb{C}$ . In particular, a nonzero unimodular Fermat triple in  $\mathbf{Oz}$  has ordinary integer coordinates.*

*Proof.* First suppose that all ratios among  $x, y, z$  are complex constants:  $x = \mu z, y = \nu z$  with  $\mu, \nu \in \mathbb{C}^\times$ . A unit-ideal witness  $ux + vy + wz = 1$  gives  $z(\mu u + \nu v + w) = 1$ , so  $z$  is a unit of  $\mathcal{B}_{\mathbb{C}}$ , and  $z$ , hence  $x, y$ , are constants by [proposition 2.5](#).

It remains to rule out a nonconstant ratio. Choose a set-sized workspace  $H$  containing the supports of  $x, y, z$  and of a unit-ideal witness; the ratios  $u = x/z$  and  $v = y/z$  lie in the Hahn field over  $H$  and satisfy  $u^n + v^n = 1$ . If one of  $u, v$  is constant, the other is algebraic over  $\mathbb{C}$ , hence equal to one of the complex roots of its polynomial, because that polynomial splits over  $\mathbb{C}$  and the Hahn field has no zero divisors. So both are nonconstant. By [lemma 7.2](#), choose  $\partial$  with  $\partial v \neq 0$ ; differentiating  $u^n + v^n = 1$  gives  $\partial u \neq 0$ .

All three Wronskians are nonzero. Indeed  $W_{\partial}(y, z) = -z^2 \partial v$  and  $W_{\partial}(z, x) = z^2 \partial u$ . If  $W_{\partial}(x, y) = 0$ , then  $\partial u/u = \partial v/v = h$ , say, and differentiating  $u^n + v^n = 1$  gives  $nh(u^n + v^n) = nh = 0$ , so  $h = 0$ , contrary to  $\partial v \neq 0$ .

Differentiating the Fermat equation gives  $x^{n-1} \partial x + y^{n-1} \partial y = z^{n-1} \partial z$ . Multiplying by  $z$  and using  $z^n = x^n + y^n$ , then multiplying by  $y$  and using  $y^n = z^n - x^n$ , gives the two relations

$$x^{n-1} W_{\partial}(z, x) = y^{n-1} W_{\partial}(y, z), \quad -x^{n-1} W_{\partial}(x, y) = z^{n-1} W_{\partial}(y, z).$$

By [lemma 7.8](#), applied in  $\mathcal{B}_{\mathbb{C}, H}$ , there are  $r, s$  with  $rx^{n-1} + sy^{n-1} = 1$ ; multiplying by  $W_{\partial}(y, z)$  and using the first relation gives  $W_{\partial}(y, z) = x^{n-1}(rW_{\partial}(y, z) + sW_{\partial}(z, x))$ . The same argument with the roles of the comaximal pairs exchanged, in both relations, gives

$$x^{n-1} \mid W_{\partial}(y, z), \quad y^{n-1} \mid W_{\partial}(z, x), \quad z^{n-1} \mid W_{\partial}(x, y) \quad (7.4)$$

in  $\mathcal{B}_{\mathbb{C}, H}$ . Let  $\alpha$  be the greatest of the three coordinate degrees and  $\beta, \gamma$  the other two. If  $\alpha = 0$ , all coordinates are constants. If  $\alpha > 0$ , the divisibility in (7.4) for the coordinate of degree  $\alpha$  and the weak Wronskian bound give

$$(n-1)\alpha \leq \beta + \gamma \leq 2\alpha.$$

For  $n > 3$  this is impossible. For  $n = 3$  it forces  $\beta = \gamma = \alpha$ , and the relevant Wronskian then has degree strictly less than  $2\alpha$  by [lemma 7.3](#), contradicting its divisibility by a nonzero element of degree  $2\alpha$ . Hence a nonconstant ratio is impossible. A unit-ideal witness in  $\mathbf{Oz}$  is also one in  $\mathcal{B}_{\mathbb{C}}$ , and constant omnific elements are ordinary integers.  $\square$

**Corollary 7.10** (An omnific form of Fermat's Last Theorem [05]). *For every ordinary integer  $n \geq 3$  there is no triple of nonzero omnific integers with  $x^n + y^n = z^n$  and  $ux + vy + wz = 1$  for some  $u, v, w \in \mathbf{Oz}$ .*

*Proof.* By [theorem 7.9](#) such a triple is ordinary, and the classical theorem of Wiles and Taylor–Wiles excludes it [15, 16]. Its positive-integer formulation also excludes nonzero signed triples: for even exponents take absolute values, and for odd exponents move negative terms to the other side.  $\square$

The rigidity theorem is the argument supplied by source 05; the final nonexistence imports the classical theorem. The unimodular Pythagorean family of [proposition 11.4](#) is infinite, so the threshold  $n \geq 3$  is sharp. Zero coordinates are excluded intentionally:  $(1, 0, 1)$  is a unimodular constant solution for every  $n$ . The theorem uses actual Bézout witnesses and actual quotient rings; it makes no assertion about triples that merely lack a common nonunit divisor ([question 14.2](#)).

## 8 The quadratic-level dichotomy

Binary Pell rigidity might suggest that a nonzero constant quadratic level never contains infinite points. This is false as soon as there is enough indefinite dimension. The following result gives a precise classification.

**Theorem 8.1** (Quadratic-level dichotomy [01]; case (iii) also [02]). *Let  $n \geq 1$ , let  $q \in \mathbb{Z}[X_1, \dots, X_n]$  be a homogeneous quadratic form, and let  $c \in \mathbb{Z} \setminus \{0\}$ . Write*

$$V_{\mathbf{O}\mathbf{Z}} = \{\mathbf{x} \in \mathbf{O}\mathbf{Z}^n : q(\mathbf{x}) = c\}, \quad V_{\mathbb{Z}} = \{\mathbf{a} \in \mathbb{Z}^n : q(\mathbf{a}) = c\}.$$

*If  $V_{\mathbb{Z}}$  is empty, so is  $V_{\mathbf{O}\mathbf{Z}}$ . Otherwise the following alternatives are exhaustive.*

- (i) *If  $q$  is nondegenerate and is either definite or of dimension at most two, then  $V_{\mathbf{O}\mathbf{Z}} = V_{\mathbb{Z}}$ .*
- (ii) *If  $q$  is degenerate, every  $\mathbf{a} \in V_{\mathbb{Z}}$  lies on an injective affine-linear family  $\mathbf{a} + t\mathbf{v}$ ,  $t \in \Pi$ , in  $V_{\mathbf{O}\mathbf{Z}}$ , with every  $t \neq 0$  giving an infinite point.*
- (iii) *If  $q$  is nondegenerate, indefinite, and  $n \geq 3$ , every  $\mathbf{a} \in V_{\mathbb{Z}}$  lies on an injective quadratic polynomial family  $\mathbf{a} + t\mathbf{v} + t^2\mathbf{w}$ ,  $t \in \Pi$ , in  $V_{\mathbf{O}\mathbf{Z}}$ , with every  $t \neq 0$  giving an infinite point. The family is injective on all of  $\mathbf{N}\mathbf{o}$ , so every such fiber contains a proper class of points.*

*In alternatives (ii) and (iii), the constant tuple of every point in the family is  $\mathbf{a}$ .*

We first prove the geometric construction needed in the indefinite case. All vectors and bilinear forms in it are ordinary real objects; only the parameter is later allowed to be surreal.

**Lemma 8.2** (A nilpotent quadratic isometry [01, 02]). *Let  $B$  be a symmetric bilinear form on  $\mathbb{R}^n$ , with  $q(x) = B(x, x)$ . Suppose  $e, u \in \mathbb{R}^n$  satisfy*

$$e \neq 0, \quad q(e) = 0, \quad B(e, u) = 0, \quad k = q(u) \neq 0,$$

*and put  $N(x) = B(x, e)u - B(x, u)e$ . Then*

$$N^2(x) = -kB(x, e)e, \quad N^3 = 0, \quad B(Nx, y) + B(x, Ny) = 0. \quad (8.1)$$

*For a parameter  $t$  in any field extension of  $\mathbb{R}$ , the polynomial map*

$$U_t = \text{id} + tN + \frac{t^2}{2}N^2, \quad U_t(x) = x + t(B(x, e)u - B(x, u)e) - \frac{k}{2}t^2B(x, e)e, \quad (8.2)$$

*satisfies*

$$q(U_t(x)) = q(x), \quad \det U_t = 1, \quad U_s U_t = U_{s+t}, \quad U_t^{-1} = U_{-t}.$$

*If  $t \in \Pi$ , all matrix entries of  $U_t$  lie in  $\mathbf{O}\mathbf{Z}$  and its entrywise constant coefficient is the identity. If  $\alpha = B(a, e) \neq 0$ , the curve  $t \mapsto U_t(a)$  is injective.*

*Proof.* From  $B(e, e) = B(e, u) = 0$  one has  $N(e) = 0$  and  $N(u) = -ke$ , which gives the first two identities of (8.1). Directly,  $B(Nx, y) = B(x, e)B(u, y) - B(x, u)B(e, y) = -B(x, Ny)$ .

For preservation, put  $\alpha = B(x, e)$  and  $\beta = B(x, u)$ . In (8.2) the added vector is  $t\alpha u - (t\beta + \frac{k}{2}t^2\alpha)e$ . Its scalar product with  $x$ , multiplied by two, is  $-kt^2\alpha^2$ , because the two linear terms cancel. Its own quadratic value is  $kt^2\alpha^2$ , since  $e$  is isotropic and orthogonal to  $u$ . Their sum is zero.

Multiplying these polynomials in  $N$  and using  $N^3 = 0$  gives  $U_s U_t = U_{s+t}$  and hence  $U_t U_{-t} = \text{id}$  over  $\mathbb{R}[t]$ . The determinant is therefore a unit of  $\mathbb{R}[t]$ , a nonzero real constant, and evaluation at  $t = 0$  makes it 1. All identities remain valid after substituting a surreal  $t$ . Every entry of  $U_t$  is an entry of  $\text{id}$  plus a real combination of  $t$  and  $t^2$ ; for  $t \in \Pi$  that difference lies in  $\Pi$ .

Finally, for  $\alpha = B(a, e) \neq 0$  and  $\beta = B(a, u)$ ,

$$U_t(a) = a + tv + t^2w, \quad v = \alpha u - \beta e, \quad w = -\frac{k\alpha}{2}e.$$

The real vectors  $u, e$  are independent (as  $q(u) \neq 0 = q(e)$ ), so  $v, w$  are independent:  $w$  is a nonzero multiple of  $e$  and  $v$  has nonzero  $u$ -component. Their independence persists over a field extension because some  $2 \times 2$  real minor is nonzero. If  $U_s(a) = U_t(a)$  with  $s \neq t$ , then  $v + (s + t)w = 0$ , a contradiction.  $\square$

Source 01 wrote this map as a transvection  $T_t$  and source 02 as the nilpotent operator  $U_T$  with  $\text{id} + TN + \frac{T^2}{2}N^2$ ; they are the same map. No infinite exponential series is used: the expression is a finite polynomial because  $N^3 = 0$ .

**Lemma 8.3** (An isotropic direction visible from a nonzero point [01, 02]). *If  $q$  is a nondegenerate indefinite real quadratic form of dimension at least three and  $a \in \mathbb{R}^n$  is nonzero, there exist  $e, u$  as in lemma 8.2 with  $B(a, e) \neq 0$ .*

*Proof.* The real isotropic vectors span  $\mathbb{R}^n$ . In a real basis in which  $q$  has positive unit vectors  $p_i$  and negative unit vectors  $n_j$  (both types exist by indefiniteness), each  $p_i \pm n_j$  is isotropic, and sums and differences recover every basis vector. Since the nonzero functional  $B(a, -)$  cannot vanish on a spanning set, some isotropic  $e$  has  $B(a, e) \neq 0$ .

Nondegeneracy gives  $f_0$  with  $B(e, f_0) = 1$ . Replacing it by  $f = f_0 - \frac{1}{2}q(f_0)e$  makes  $q(f) = 0$  while retaining  $B(e, f) = 1$ . The plane spanned by  $e, f$  is nondegenerate, so its orthogonal complement is nondegenerate of dimension  $n - 2 \geq 1$  and contains  $u$  with  $q(u) \neq 0$  (otherwise polarization would make its form zero). This  $u$  is orthogonal to  $e$ .  $\square$

Source 01 stated this lemma for  $q(a) \neq 0$ ; its proof, like source 02's, uses only  $a \neq 0$ .

*Proof of theorem 8.1.* The empty-set assertion is theorem 5.1.

Suppose first that  $q$  is definite and  $\mathbf{x} \in V_{\mathbf{Oz}}$  has an infinite coordinate. Let  $\gamma > 0$  be the largest leading exponent of its nonzero coordinates and  $r_i$  the coefficient of  $\omega^\gamma$  in  $x_i$  (zero when absent). The real vector  $\mathbf{r}$  is nonzero, and the coefficient of  $\omega^{2\gamma}$  in  $q(\mathbf{x})$  is exactly  $q(\mathbf{r}) \neq 0$ , contradicting  $q(\mathbf{x}) = c$ . Thus every coordinate is finite and hence an ordinary integer. In dimension two, nondegeneracy means that the binary quadratic has two distinct linear factors over  $\mathbb{C}$ , so theorem 6.3 applies. A nondegenerate one-dimensional form is definite. This proves (i).

For (ii), choose a nonzero real vector  $\mathbf{v} \in \text{rad}(B)$ . For every  $t$ ,  $q(\mathbf{a} + t\mathbf{v}) = q(\mathbf{a}) = c$ . If  $t \in \Pi$ , the coordinates of  $t\mathbf{v}$  lie in  $\Pi$ , so  $\mathbf{a} + t\mathbf{v} \in \mathbf{Oz}^n$  with constant tuple  $\mathbf{a}$ . The map is injective because  $\mathbf{v} \neq 0$ , and for  $t \neq 0$  some coordinate has positive degree.

For (iii),  $\mathbf{a} \neq 0$  since  $c \neq 0$ . Choose  $e, u$  by lemma 8.3 and use lemma 8.2. Each coordinate of  $U_t(\mathbf{a})$  is an integer constant plus real multiples of  $t$  and  $t^2$ , so it lies in  $\mathbf{Oz}$  with constant coefficient  $a_i$  for  $t \in \Pi$ . Injectivity was proved for all surreal  $t$ . If  $t \neq 0$ , its degree is positive, and in a coordinate where  $w_i \neq 0$  the quadratic term has degree  $2 \deg_\omega(t)$ , strictly larger than that of the linear term, so the point is infinite. Since  $\Pi$  contains  $\omega^\gamma$  for every surreal  $\gamma > 0$ , the fiber contains a proper class of points.  $\square$

### 8.1 Explicit families

For positive ordinary integers  $D, D'$  and  $t \in \Pi$ , set [01]

$$x = 1 + \frac{D'}{2}t^2, \quad y = \frac{D'}{2\sqrt{D}}t^2, \quad z = t. \quad (8.3)$$

All three coordinates are omnific integers, and  $x^2 - Dy^2 - D'z^2 = 1$ : the two quartic terms cancel, as do the two quadratic terms. For  $t \neq 0$  every coordinate is infinite. The tuple is unimodular, since

$$1 = x - \left(\frac{D'}{2}t\right)z, \quad \frac{D'}{2}t \in \mathbf{Oz}. \quad (8.4)$$

This illustrates a sharp contrast:

$$\begin{aligned} x^2 - Dy^2 &= 1 \text{ has only ordinary points,} \\ x^2 - Dy^2 - D'z^2 &= 1 \text{ has infinite unimodular points.} \end{aligned}$$

There is no appeal to convergence, analytic continuation, or infinite iteration; every expression is a finite polynomial.

For the Lorentz form  $q(x, y, z) = x^2 - y^2 - z^2$  with Gram matrix  $G = \text{diag}(1, -1, -1)$ , take  $e = (1, 1, 0)$  and  $u = (0, 0, 1)$  [02]. Then  $q(u) = -1$ ,  $N(x, y, z) = (z, z, x - y)$ , and

$$U_t = \begin{pmatrix} 1 + t^2/2 & -t^2/2 & t \\ t^2/2 & 1 - t^2/2 & t \\ t & -t & 1 \end{pmatrix}, \quad U_t^\top G U_t = G, \quad \det U_t = 1, \quad U_s U_t = U_{s+t}. \quad (8.5)$$

For  $t \in \Pi$  this is a one-parameter family of omnific Lorentz matrices with constant coefficient the identity. Acting on  $(1, 0, 0)$  gives  $(1 + t^2/2, t^2/2, t)$ , and replacing  $t$  by  $2s$  gives the denominator-free identity

$$(1 + 2s^2)^2 - (2s^2)^2 - (2s)^2 = 1,$$

valid for every  $s \in \mathbf{Oz}$ , with constant fiber  $(1, 0, 0)$  whenever  $s \in \Pi$ . For  $s = \omega$  the point is  $(1 + 2\omega^2, 2\omega^2, 2\omega)$ ; the squares contain terms of degrees four, two and zero, and the nonconstant terms cancel exactly. There is no negative-exponent tail. This is concretely how an extra negative square evades the Pell rigidity mechanism.

**Corollary 8.4** (An invisible unipotent subgroup [02]). *For every nondegenerate indefinite integral quadratic form  $q$  in at least three variables, there is an injective group homomorphism*

$$(\Pi, +) \longrightarrow \ker(\text{SO}(q, \mathbf{Oz}) \xrightarrow{\text{ct}} \text{SO}(q, \mathbb{Z})), \quad t \longmapsto U_t.$$

*Proof.* Choose a nonzero real test vector  $a$  and use lemma 8.3 to construct  $e, u$ . Lemma 8.2 gives a homomorphism with entries in  $\mathbf{Oz}$ , determinant one, preservation of  $q$ , and constant coefficient id. If  $U_s = U_t$ , apply both to  $a$ ; injectivity of  $t \mapsto U_t(a)$  gives  $s = t$ . No ordinary point on a prescribed level is needed.  $\square$

Here  $\mathrm{SO}(q, \mathbf{Oz})$  means determinant-one matrices over  $\mathbf{Oz}$  preserving the integral quadratic polynomial  $q$ ; entrywise it lands in  $\mathrm{SO}(q, \mathbb{Z})$ . This subgroup is invisible to every finite quotient of the coefficient ring ([corollary 3.4](#)), yet it is a proper class. It concerns indefinite forms only; for definite forms see [corollary 9.4](#).

## 8.2 What the classification does and does not say

The classification concerns the presence of genuinely new points on a *represented nonzero level* [01, 02]. It does not decide whether the integer level is represented in the first place. Nor does it classify all infinite points or claim that every omnific point of a higher-dimensional quadric lies on the constructed orbit: a displayed curve is a rigorously constructed subset of the full fiber. Source 02's version covers nondegenerate forms only and excludes degenerate forms and the zero level; source 01's alternative (ii) covers degenerate forms.

For zero levels, homogeneity gives a different mechanism. An indefinite real quadratic form has real isotropic directions, and multiplying any such direction by an element of  $\Pi$  produces omnific points with zero constant tuple. Such points can exist even when there is no nonzero ordinary integer isotropic vector.

## 9 Bounded geometry, leading forms, and orthogonal matrices

Multiplicative rigidity is not the only mechanism. A different one comes from boundedness and the discreteness of  $\mathbf{Oz}$ .

### 9.1 Bounded sets

**Proposition 9.1** (Positive bounds [01, 05]). *If  $a_1, \dots, a_n$  are positive ordinary reals,  $m_1, \dots, m_n \geq 1$  are ordinary integers,  $x_1, \dots, x_n \in \mathbf{Oz}$ , and*

$$\sum_i a_i x_i^{2m_i} = C \in \mathbb{R},$$

*then every  $x_i$  is an ordinary integer. In particular, an omnific point on a sphere of ordinary finite radius is an ordinary integer point.*

*Proof.* Each summand is nonnegative and at most  $C$ . If  $C < 0$  there is no solution; otherwise this gives an ordinary real bound on each  $|x_i|$ , and [proposition 2.5](#) applies.  $\square$

**Theorem 9.2** (Bounded semialgebraic rigidity [02]). *Let  $X \subseteq \mathbb{R}^m$  be a bounded semialgebraic set defined by a finite formula with real coefficients, and let  $X(\mathbf{No})$  be the class defined by the same formula in  $\mathbf{No}$ . Then  $X(\mathbf{No}) \cap \mathbf{Oz}^m = X \cap \mathbb{Z}^m$ .*

*Proof.* Choose an ordinary real  $M > 0$  with  $|x_i| \leq M$  for all  $\mathbf{x} \in X$ . This universal implication is a first-order sentence over  $\mathbb{R}$ . Given a candidate point of  $X(\mathbf{No})$ , generate a field from its coordinates and  $\mathbb{R}$  and take its real closure inside  $\mathbf{No}$ ; this is a set-sized real closed subfield. Tarski's quantifier elimination makes the inclusion of  $\mathbb{R}$  into it elementary [11], so the bound transfers and every omnific coordinate is bounded by  $M$ , hence an ordinary integer by [proposition 2.5](#). An integer tuple satisfies the formula in the extension exactly when it does over  $\mathbb{R}$ , again by elementarity.  $\square$

**Corollary 9.3** (Definite levels [01, 02]). *For a positive or negative definite real quadratic form  $q$  and a real  $c$ , every omnific point of  $q(\mathbf{x}) = c$  is an ordinary integer tuple.*

*Proof.* A nonempty real level set of a definite form is bounded, and [theorem 9.2](#) applies. Alternatively, the leading-coefficient argument in the proof of [theorem 8.1\(i\)](#) applies verbatim to any real level  $c$  [01].  $\square$

For instance,  $x^2 + y^2 = 1$  has exactly  $(\pm 1, 0), (0, \pm 1)$  as omnific solutions, and so has  $x^4 + y^4 = 1$ . These concern a fixed finite level; the homogeneous Pythagorean relation lets all coordinates grow together. Positivity does *not* imply being a sum of squares in  $\mathbf{Oz}$  ([remark 10.8](#)).

## 9.2 Orthogonal and symmetric matrices

**Corollary 9.4** (Euclidean orthogonal group [01, 02]). *For every ordinary  $n \geq 1$ ,  $O_n(\mathbf{Oz}) = O_n(\mathbb{Z})$  is the group of signed permutation matrices, of order  $2^n n!$ , and  $SO_n(\mathbf{Oz})$  consists of the signed permutation matrices of determinant one.*

*Proof.* For an orthogonal matrix, the sum of the squares in each row is 1, so every entry is ordinary by [proposition 9.1](#). An integer row of squared norm one has exactly one nonzero entry, equal to  $\pm 1$ , and orthogonality of different rows forces those entries into different columns. Conversely each signed permutation matrix is orthogonal.  $\square$

Requiring *surreal* entries gives the large rotation group of a real closed field, whereas *omnific* entries in standard Euclidean coordinates leave a finite group. This is not a claim that matrix groups over  $\mathbf{Oz}$  are finite:  $SL_2(\mathbf{Oz})$  contains all  $\begin{pmatrix} 1 & t \\ 0 & 1 \end{pmatrix}$ ,  $t \in \mathbf{Oz}$ , and indefinite forms have the unipotent subgroup of [corollary 8.4](#). The Euclidean group is rigid because of positivity.

**Corollary 9.5** (Symmetric matrix rigidity [01]). *Let  $M$  be a symmetric matrix over  $\mathbf{Oz}$ . If  $\text{tr}(M^2)$  is an ordinary real constant, every entry of  $M$  is an ordinary integer. In particular, this holds if its characteristic polynomial has ordinary integer coefficients.*

*Proof.* For symmetric  $M$ ,  $\text{tr}(M^2) = \sum_i m_{ii}^2 + 2 \sum_{i < j} m_{ij}^2$ , so [proposition 9.1](#) applies. When the size is at least two,  $\text{tr}(M^2) = (\text{tr } M)^2 - 2e_2(M)$ , where  $e_2(M)$ , the sum of the principal  $2 \times 2$  minors, is up to sign a coefficient of the characteristic polynomial, as is  $\text{tr } M$ ; for size one the assertion is immediate.  $\square$

## 9.3 No real point at infinity

**Theorem 9.6** (Leading-homogeneous obstruction [01]). *Let  $F_1, \dots, F_r \in \mathbb{Z}[X_1, \dots, X_n]$  be nonconstant polynomials, and let  $H_j$  be the top homogeneous part of  $F_j$ . If the  $H_j$  have no common nonzero real zero, then every common zero of the  $F_j$  in  $\mathbf{Oz}^n$  lies in  $\mathbb{Z}^n$ .*

*Proof.* Suppose a common zero  $\mathbf{x}$  has an infinite coordinate. Let  $\gamma > 0$  be the largest leading exponent of its coordinates and  $\mathbf{r} \neq 0$  the real vector of coefficients of  $\omega^\gamma$ . If  $d_j = \deg F_j$ , the coefficient at  $\omega^{d_j \gamma}$  in  $F_j(\mathbf{x})$  is  $H_j(\mathbf{r})$ : a product attains that exponent only by taking exponent  $\gamma$  in each of  $d_j$  factors, and terms of lower ordinary total degree have exponent strictly less than  $d_j \gamma$ . So every  $H_j(\mathbf{r}) = 0$ , contrary to the hypothesis. All coordinates are therefore finite and hence ordinary integers.  $\square$

The hypothesis is a sufficient absence-of-real-directions-at-infinity condition for the chosen system. Its converse is false: a Pell equation has real directions at infinity when  $D > 0$ , but its nonzero fibers are rigid by factorization. Initial-form obstructions and constant-product obstructions detect different mechanisms.

## 10 Defining the ordinary integers

The retraction  $\text{ct}$  detects ordinary coefficients but is not a symbol of the ring language. Pell rigidity shows that the embedded copy of  $\mathbb{Z}$  is nevertheless detected by a single polynomial equation with existential auxiliary variables. Two versions were found independently.

### 10.1 Pell coordinates as finite bounds

Define ordinary Pell solutions by

$$X_k + Y_k\sqrt{2} = (3 + 2\sqrt{2})^k, \quad (X_{k+1}, Y_{k+1}) = (3X_k + 4Y_k, 2X_k + 3Y_k), \quad k \geq 0, \quad (10.1)$$

with  $(X_0, Y_0) = (1, 0)$ .

**Lemma 10.1** (Pell witnesses [01, 05]). *For every ordinary  $k \geq 0$ ,  $X_k^2 - 2Y_k^2 = 1$ ; for  $k \geq 1$  both coordinates are positive and  $X_k \geq 3^k$ . Moreover  $X_{2j} \equiv 1$  and  $X_{2j+1} \equiv 3 \pmod{8}$ , so both parity subsequences are unbounded.*

*Proof.* Direct expansion gives  $(3X + 4Y)^2 - 2(2X + 3Y)^2 = X^2 - 2Y^2$ , and ordinary induction proves the identity, positivity and  $X_{k+1} \geq 3X_k$ . From  $X_k = \frac{1}{2}((3 + 2\sqrt{2})^k + (3 - 2\sqrt{2})^k)$  one gets the recurrence  $X_{k+2} = 6X_{k+1} - X_k$  with  $X_0 = 1, X_1 = 3$ ; modulo 8 it gives the alternating residues  $1, 3, 1, 3, \dots$  (as  $6 \cdot 3 - 1 \equiv 1$  and  $6 \cdot 1 - 3 \equiv 3$ ).  $\square$

### 10.2 The five-auxiliary-variable guard

We use the Legendre–Gauss three-square theorem: a nonnegative ordinary integer  $N$  is a sum of three integer squares if and only if it is not of the form  $4^a(8b + 7)$  with  $a, b \geq 0$ . It is imported, not reproved; see the exposition [12] and the formal development in Isabelle’s Archive of Formal Proofs [13].

**Lemma 10.2** (A three-square gap below a Pell coordinate [01]). *For every ordinary nonnegative integer  $S$ , there are ordinary integers  $u, v, w_1, w_2, w_3$  such that*

$$u^2 - 2v^2 = 1, \quad u = S + w_1^2 + w_2^2 + w_3^2.$$

*Proof.* Choose the residue of  $u$  modulo 8 according to the following table; each choice is realized by an unbounded subsequence of the  $X_k$  (lemma 10.1).

| $S \bmod 8$        | 0 | 1 | 2 | 3 | 4 | 5 | 6 | 7 |
|--------------------|---|---|---|---|---|---|---|---|
| chosen $u \bmod 8$ | 1 | 3 | 3 | 1 | 1 | 3 | 1 | 1 |
| $(u - S) \bmod 8$  | 1 | 2 | 1 | 6 | 5 | 6 | 3 | 2 |

Take  $u$  in that subsequence so large that  $N = u - S > 0$ . Its residue is one of  $1, 2, 3, 5, 6$ , so it is neither divisible by 4 nor congruent to 7 modulo 8, and cannot be  $4^a(8b + 7)$ . The three-square theorem supplies the  $w_j$ , and the corresponding Pell coordinate supplies  $v$ .  $\square$

For each ordinary  $n \geq 1$ , define the integer polynomial

$$\Psi_n(\mathbf{X}, U, V, \mathbf{W}) = (U^2 - 2V^2 - 1)^2 + \left(U - \sum_{i=1}^n X_i^2 - W_1^2 - W_2^2 - W_3^2\right)^2, \quad (10.2)$$

of total degree four in  $n + 5$  variables.

**Theorem 10.3** (Uniform quartic integrality guard [01]). *For every  $\mathbf{x} \in \mathbf{Oz}^n$ ,*

$$\mathbf{x} \in \mathbb{Z}^n \iff \exists u, v, w_1, w_2, w_3 \in \mathbf{Oz} \quad \Psi_n(\mathbf{x}, u, v, w_1, w_2, w_3) = 0. \quad (10.3)$$

*Every coordinate of every zero of  $\Psi_n$  in  $\mathbf{Oz}^{n+5}$  is an ordinary integer. The number of auxiliary variables is five, independently of  $n$ .*

*Proof.* Suppose  $\Psi_n = 0$  in  $\mathbf{Oz}$ . A sum of two squares in an ordered ring vanishes only when both squares vanish, so

$$u^2 - 2v^2 = 1, \quad u = \sum_i x_i^2 + w_1^2 + w_2^2 + w_3^2. \quad (10.4)$$

By Pell rigidity (corollary 6.5),  $u, v \in \mathbb{Z}$ . The second equation gives  $u \geq 0$  and bounds the square of each  $x_i$  and  $w_j$  by the ordinary number  $u$ , so each is an ordinary integer by proposition 2.5. Conversely, for  $\mathbf{x} \in \mathbb{Z}^n$  put  $S = \sum_i x_i^2$  and choose the witnesses of lemma 10.2.  $\square$

For  $x = 1$ , one witness is  $u = 3, v = 2, (w_1, w_2, w_3) = (1, 1, 0)$ ; for  $x = 2$ ,  $u = 17, v = 12, (w_1, w_2, w_3) = (3, 2, 0)$ . The assertions for all integers rely on lemma 10.2, not on examples or a finite computation.

### 10.3 The four-square definition

**Theorem 10.4** (A quartic definition of  $\mathbb{Z}$  [05]). *For  $t \in \mathbf{Oz}$ , the following are equivalent:*

- (i)  $t \in \mathbb{Z}$ ;
- (ii) *there exist  $x, y, z_1, z_2, z_3, z_4 \in \mathbf{Oz}$  with*

$$x^2 - 2y^2 = 1, \quad x - t^2 = z_1^2 + z_2^2 + z_3^2 + z_4^2; \quad (10.5)$$

- (iii) *the quartic equation*

$$(x^2 - 2y^2 - 1)^2 + \left(x - t^2 - \sum_{j=1}^4 z_j^2\right)^2 = 0 \quad (10.6)$$

*has a solution in its six existential variables over  $\mathbf{Oz}$ .*

*Every witness to (10.5) has all six coordinates in  $\mathbb{Z}$ .*

*Proof.* If (10.5) holds, Pell rigidity makes  $x, y$  ordinary, and  $0 \leq t^2 \leq x$  bounds  $t$ , so  $t \in \mathbb{Z}$ ; each  $z_j^2$  is bounded by the ordinary integer  $x - t^2$ , so every  $z_j$  is ordinary. Conversely, for  $t \in \mathbb{Z}$  choose  $X_k \geq t^2$  by lemma 10.1; the nonnegative integer  $X_k - t^2$  is a sum of four squares by Lagrange's theorem, which is available in Mathlib as `Nat.sum_four_squares` [14]. In an ordered ring a sum of two squares vanishes exactly when both do, so (10.5) and (10.6) are equivalent. The polynomial has total degree four, integer coefficients and no infinite parameters.  $\square$

Write  $\text{Std}(t)$  for the existential formula (10.6) and  $\text{Std}_n(\mathbf{x})$  for (10.3). Replacing  $t^2$  by  $\sum_{i=1}^n t_i^2$  in (10.5) defines  $\mathbb{Z}^n$  with six auxiliary variables by the same proof. Here “standard” refers precisely to the embedded  $\mathbb{Z}$ ; it is not an extra predicate of the language.

*Remark 10.5* (Comparison of the two guards). Source 01's guard is uniform in  $n$  with five auxiliary variables and needs the three-square theorem; source 05's uses one more auxiliary variable and Lagrange's four-square theorem, and source 01 had noted this simpler variant without writing it out. Neither source claims its count optimal ([question 14.5](#)). For formalization the difference matters: at the Mathlib revision pinned by this repository (81a5d257), `Nat.sum_four_squares` is proved in `Mathlib/NumberTheory/SumFourSquares.lean`, while no three-square theorem is present. Only source 05's definition therefore rests on a theorem in Mathlib; source 01's would need the three-square theorem imported or proved first. The four-square theorem is used only on an ordinary nonnegative integer; its analogue for positive omnific integers is false ([remark 10.8](#)).

## 10.4 Definability consequences

$\text{Std}(\mathbf{Oz}) = \mathbb{Z}$ : ordinary integer arithmetic is present not only as the quotient  $\mathbf{Oz}/\Pi$  but as an existentially definable subring of  $\mathbf{Oz}$ . There is no conflict with [proposition 5.3](#), because  $\mathbb{Z}$  is stable under `ct`, whereas the positive cone and the nonzero locus are not.

**Corollary 10.6** (Definability of `ct` and  $\Pi$  [01, 05]). *The graph of `ct` and the ideal  $\Pi$  are first-order definable in the pure ring language of  $\mathbf{Oz}$  without parameters:*

$$\begin{aligned} \text{ct}(a) = m &\iff \text{Std}(m) \wedge \forall k ((\text{Std}(k) \wedge k \neq 0) \Rightarrow \exists y (a - m = ky)), \\ j \in \Pi &\iff \forall k ((\text{Std}(k) \wedge k \neq 0) \Rightarrow \exists y (j = ky)). \end{aligned}$$

*Proof.* The right sides say that  $m \in \mathbb{Z}$  and that  $a - m$  (respectively  $j$ ) is divisible by every nonzero ordinary integer; by [\(3.2\)](#) this means  $a - m \in \Pi$  (respectively  $j \in \Pi$ ), which for  $m \in \mathbb{Z}$  is equivalent to  $\text{ct}(a) = m$ .  $\square$

These definitions use a universal quantifier and an implication. Neither source claims an existential (Diophantine) definition of  $\Pi$  or of the graph of `ct` ([question 14.4](#)).

*Remark 10.7* (An abstract principle [01]). The proof of [theorem 10.3](#) needs less than all surreal numbers. Let  $R$  be any discretely ordered ring containing  $\mathbb{Z}$  in which every solution of  $U^2 - 2V^2 = 1$  is ordinary. Every element of  $R$  bounded by an ordinary integer is ordinary, by discreteness and a finite comparison with the integers in the bounding interval; so the same proof shows that  $\Psi_n$  defines  $\mathbb{Z}^n$  in  $R^n$ , and likewise for [\(10.6\)](#). The omnific rings considered here satisfy the Pell hypothesis by [corollary 6.5](#).

## 10.5 Sums of squares and induction

*Remark 10.8* (Squares and Peano arithmetic [01, 02, 05]). The positive omnific integer  $\omega - 1$  is not a sum of any finite number of squares in  $\mathbf{Oz}$ : applying `ct` to such an identity gives  $-1 = \sum_j m_j^2$  with  $m_j \in \mathbb{Z}$ . So the nonnegative cone of  $\mathbf{Oz}$  does not satisfy Peano arithmetic, in which Lagrange's theorem is provable. This is compatible with the classical fact that integer parts of real closed fields satisfy open induction [6, 3]; open induction is not full arithmetic.

**Theorem 10.9** (A failed existential induction instance [05]). *On the nonnegative part of  $\mathbf{Oz}$ , consider the formula*

$$\Phi(t) : \exists x, y \geq 0 (x^2 = 2y^2 + 1 \wedge x > t \wedge y > 0), \quad (10.7)$$

*existential with a quantifier-free ordered-ring matrix. Over  $\mathbf{Oz}_{\geq 0}$ ,*

$$\Phi(0), \quad \forall t \geq 0 (\Phi(t) \Rightarrow \Phi(t+1)), \quad \neg \Phi(\omega),$$

*and  $\Phi(t)$  holds exactly for the ordinary nonnegative integers  $t$ .*

*Proof.* The pair  $(3, 2)$  witnesses  $\Phi(0)$ . Given witnesses  $x, y$  for  $\Phi(t)$ , set  $x' = 3x + 4y$ ,  $y' = 2x + 3y$ . The identity of lemma 10.1 gives  $(x')^2 = 2(y')^2 + 1$ ; positivity and discreteness give  $x, y \geq 1$ , so  $x' > x + 1 > t + 1$  and  $y' > 0$ . This proves the successor step for every omnific  $t$ . Pell rigidity makes every witness ordinary, so  $x$  cannot exceed an infinite  $t$  such as  $\omega$ . Conversely an ordinary nonnegative  $t$  has a larger positive ordinary Pell witness by lemma 10.1. By proposition 2.5 these are the only two kinds of nonnegative omnific integers.  $\square$

By Shepherdson's theorem, an integer part of a real closed ordered field satisfies open induction, that is, induction for quantifier-free formulas [6]. In a divisible set-sized Hahn workspace  $H$ , the ring  $\mathbf{Oz}_H$  is an integer part of the real closed field of real Hahn series with exponents in  $H$  (theorem 2.7); so the full  $\mathbf{Oz}$  satisfies the same schema, by applying the theorem in a workspace containing the finitely many parameters and any proposed counterexample [05]. There is no conflict with theorem 10.9:  $\Phi$  is existential, not quantifier free. The sentence "every nonnegative  $t$  has a larger positive Pell solution" is true over  $\mathbb{Z}$  and false over  $\mathbf{Oz}$ ; thus  $\mathbf{Oz}$  is not an elementary extension of  $\mathbb{Z}$  as an ordered ring.

*Remark 10.10* (An induction failure in the pure ring language [merge]). Existential induction also fails in the pure ring language, without order: the existential formula  $\text{Std}(t)$  of theorem 10.4 satisfies  $\text{Std}(0)$  and  $\forall t (\text{Std}(t) \Rightarrow \text{Std}(t + 1))$  over  $\mathbf{Oz}$  but not  $\text{Std}(\omega)$ , because it defines  $\mathbb{Z}$ , which contains 0 and is closed under  $t \mapsto t + 1$ , and  $\omega \notin \mathbb{Z}$ . In particular the formula  $\text{Std}(t)$  also shows that  $\mathbf{Oz}$  is not an elementary extension of  $\mathbb{Z}$  in the ring language: the sentence  $\forall t \text{Std}(t)$  holds in  $\mathbb{Z}$  and fails in  $\mathbf{Oz}$ .

## 11 Infinite families, primitivity, and projective points

### 11.1 Polynomial curves through integer points

**Proposition 11.1** (Finite polynomial lifting [01, 02, 05]). *Let  $F_1, \dots, F_r \in \mathbb{R}[X_1, \dots, X_n]$ , and let  $P_1(T), \dots, P_n(T) \in \mathbb{R}[T]$  satisfy*

$$F_j(P_1(T), \dots, P_n(T)) = 0 \quad (1 \leq j \leq r), \quad P_i(0) \in \mathbb{Z} \quad (1 \leq i \leq n).$$

*Then  $\mathbf{P}(t) \in \mathbf{Oz}^n$  solves the system for every  $t \in \Pi$ , with constant tuple  $\mathbf{P}(0)$ . If some  $P_i$  is nonconstant, every  $t \in \Pi \setminus \{0\}$  produces an infinite point.*

*Proof.* Every nonconstant term of  $P_i(t)$  lies in  $\Pi$ , since  $\Pi$  is an ideal and a real vector space, so the constant coefficient is exactly  $P_i(0)$ . Polynomial identities hold under substitution into any field extension. If  $P_i$  has degree  $d > 0$  and  $\deg_\omega(t) = \gamma > 0$ , its top term has degree  $d\gamma$ , strictly larger than every other term's degree.  $\square$

No integrality is required of the nonconstant coefficients: for example  $(1 + t^2/2, t^2/2, t)$  is omnific for every  $t \in \Pi$  although  $1/2 \notin \mathbf{Oz}$ . The finiteness of the polynomials matters, however. Evaluating an infinite real power series at an infinite omnific argument is not licensed by this proposition; the Taylor expansion of  $\sqrt{1 + T}$  is not an admissible way to assert that  $\sqrt{1 + \omega}$  lies in  $\mathbf{Oz}$ . Polynomial maps supply a systematic sufficient condition for nontrivial fibers; they do not exhaust all normal-form constructions, and no such exhaustion is assumed (but see theorem 11.7 for finite supports).

### 11.2 Infinite solutions of homogeneous equations

**Theorem 11.2** (Homogeneous cone criterion [01]; part (b) [02]). *Let  $F_1, \dots, F_r \in \mathbb{R}[X_1, \dots, X_n]$  be homogeneous of positive degrees.*

- (a) *Their common zero locus has a nonzero point in  $\mathbf{Oz}^n$  if and only if it has a nonzero point in  $\mathbb{R}^n$ .*
- (b) *If the  $F_j$  have ordinary integer coefficients, the equivalence in (a) persists with any fixed combination of coordinate sign and nonvanishing conditions preserved by positive scaling, and these existence problems are decidable.*

*Proof.* (a) A real nonzero common zero  $\mathbf{a}$  gives the omnific common zero  $\omega\mathbf{a} \in \Pi^n$  by homogeneity. Conversely, take a nonzero common zero  $\mathbf{x} \in \mathbf{Oz}^n$ . If every coordinate is constant, it is a real point. Otherwise let  $\gamma > 0$  be the maximum of the coordinates' leading degrees and  $r_i = [\omega^\gamma]x_i$ ; the vector  $\mathbf{r}$  is nonzero. For  $F_j$  homogeneous of degree  $d_j$ , the coefficient at  $\omega^{d_j\gamma}$  in  $F_j(\mathbf{x})$  is  $F_j(\mathbf{r})$ , since a product attains that exponent only by taking exponent  $\gamma$  in every factor. So  $\mathbf{r}$  is a nonzero real common zero. This route uses no quantifier elimination.

(b) A real solution  $\mathbf{r}$  gives  $\omega\mathbf{r}$ , which has the same signs and zero pattern. Conversely an omnific solution lies, with  $\mathbb{R}$ , in a set-sized real closed subfield of  $\mathbf{No}$  (generate a field and take its real closure inside  $\mathbf{No}$ ). Quantifier elimination makes the inclusion of  $\mathbb{R}$  elementary [11], so the existential statement with its finite sign conditions already holds over  $\mathbb{R}$ ; quantifier elimination is an effective decision procedure for this finite formula with integer coefficients.  $\square$

The leading-vector route of (a) does not preserve sign or nonvanishing conditions on coordinates of lower degree, which is why (b) uses Tarski. Any homogeneous solution over  $\mathbf{No}$ , not only over  $\mathbb{R}$ , can be multiplied by a common positive monomial into  $\Pi^n$  preserving signs and zero patterns (corollary 4.7) [02, 05]. Neither statement decides affine equations: homogenizing an affine equation and requiring the new coordinate to be nonzero allows division in  $\mathbf{No}$ , not in  $\mathbf{Oz}$ .

**Example 11.3** (Fermat scaling [01, 02, 05]). For every ordinary integer  $n \geq 2$  and  $0 < t \in \Pi$ ,

$$x = t, \quad y = t, \quad z = 2^{1/n}t \tag{11.1}$$

is a positive infinite solution of  $x^n + y^n = z^n$ ; for  $t = \omega$  this is  $\omega^n + \omega^n = (2^{1/n}\omega)^n$ . The irrational coefficient is permitted by the definition of  $\mathbf{Oz}$ . All constant coefficients are zero, consistent with theorem 5.1, whose specialization is  $(0, 0, 0)$ ; it does not refute Fermat's Last Theorem about nonzero ordinary integers. The coordinates are divisible by every ordinary integer and have a common infinite monomial divisor (theorem 4.2), so these solutions are not primitive and must not be cited as resolving a primitive Fermat problem.

*Provenance.* Source 02 states that the failure of the nonzero analogue of Fermat's Last Theorem for omnific integers "is already known", citing comments by Emil Jeřábek (27 June 2018) and Timothy Chow (16 November 2021) under MathOverflow answer 303688 [17]. This attribution could not be verified when this report was assembled (the site was not reachable), and it is recorded as source 02's, unverified. No source claims novelty for the example.

### 11.3 Unimodular Pythagorean triples

**Proposition 11.4** (Infinite unimodular Pythagorean triples [01, 05]). *For every  $t \in \Pi$ , put  $x = t^2 - 1$ ,  $y = 2t$ ,  $z = t^2 + 1$ . Then  $x^2 + y^2 = z^2$  and the triple is unimodular. For  $t \neq 0$  all three coordinates are infinite, and for  $t > 0$  they are positive.*

*Proof.* The identity is the ordinary polynomial identity  $(t^2 - 1)^2 + 4t^2 = (t^2 + 1)^2$ . Two unit-ideal witnesses are

$$\frac{t^2}{2}z - \left(\frac{t^2}{2} + 1\right)x = 1 \text{ [01]}, \quad \frac{t}{2}y - x = 1 \text{ [05]}, \quad (11.2)$$

with  $t^2/2, t/2 \in \Pi \subseteq \mathbf{Oz}$ . Positive leading degree gives the assertion about infinite size.  $\square$

The constant tuple is  $(-1, 0, 1)$ , not the zero tuple; that is exactly where the common-divisor obstruction for  $\Pi^3$  stops applying. Infinite solutions therefore need not arise by scaling an ordinary tuple, and exponent two differs from exponents at least three (theorem 7.9). More generally  $(u^2 - v^2)^2 + (2uv)^2 = (u^2 + v^2)^2$  for all  $u, v \in \mathbf{Oz}$  [02].

#### 11.4 Which real projective directions have primitive representatives?

A point  $[r_0 : \dots : r_n] \in \mathbb{P}^n(\mathbb{R})$  is *rational* if it has homogeneous coordinates in  $\mathbb{Q}$ , equivalently a nonzero integer coordinate tuple.

**Theorem 11.5** (Rationality of primitive constant directions [01]). *For a real projective point  $p = [r_0 : \dots : r_n]$ , the following are equivalent:*

- (i)  $p$  is rational;
- (ii)  $p$  has a unimodular representative in  $\mathbf{Oz}^{n+1}$ ;
- (iii)  $p$  has a primitive representative in  $\mathbf{Oz}^{n+1}$ .

*Proof.* For (i) $\Rightarrow$ (ii), choose an integer representative with ordinary gcd one; the integer Bézout identity makes it unimodular over  $\mathbf{Oz}$ . (ii) $\Rightarrow$ (iii) is immediate. For (iii) $\Rightarrow$ (i), write a primitive representative as  $z_i = \lambda r_i$  with  $\lambda \in \mathbf{No}^\times$  and choose  $r_j \neq 0$ . Since  $z_j \in \mathbf{Oz} \subseteq \mathcal{B}_{\mathbb{R}}$  and division by the nonzero real  $r_j$  preserves  $\mathcal{B}_{\mathbb{R}}$ ,  $\lambda \in \mathcal{B}_{\mathbb{R}}$ . Put  $c = \text{ct}(\lambda)$ ; then  $\text{ct}(z_i) = cr_i \in \mathbb{Z}$  for all  $i$ . If  $c = 0$ , every  $z_i$  lies in  $\Pi$ , so 2 divides every coordinate, contradicting primitivity. Hence  $c \neq 0$  and  $(cr_0, \dots, cr_n)$  is a nonzero integer representative of  $p$ .  $\square$

For example,  $[1 : \sqrt{2}]$  has omnific representatives such as  $(\omega, \sqrt{2}\omega)$  but no primitive, hence no unimodular, representative; source 02 proved the unimodular half of this example directly. The theorem concerns real projective points; it is not a classification of all points of  $\mathbb{P}^n(\mathbf{No})$ , and neither monomial clearing nor the existence of a homogeneous solution answers the primitive-representative question.

#### 11.5 Finite supports specialize to polynomial arcs

The rigidity theorems above do not depend on finite supports. Finite-support expressions nevertheless admit an additional reduction [05].

**Lemma 11.6** (Finite ordered specialization [05]). *Let  $H$  be a finitely generated subgroup of an ordered abelian group and  $E \subseteq H$  finite. There is an additive map  $\ell : H \rightarrow \mathbb{Z}$  with  $\ell(a) > \ell(b)$  whenever  $a, b \in E$  and  $a > b$ .*

*Proof.* A finitely generated torsion-free abelian group is isomorphic to  $\mathbb{Z}^r$ . In such coordinates, form the finite set of vectors  $a - b$  for  $a, b \in E$  with  $a > b$ ; they are positive in the original order. Zero is not in their convex hull in  $\mathbb{R}^r$ : otherwise, since the vectors are integral, the rational linear system describing such a convex combination would have a rational solution (a

nonempty polytope given by rational equations, nonnegativity and a coordinate-sum equation has a rational vertex, obtained by setting a maximal set of coordinates to zero and solving the remaining independent rational equations), and clearing denominators would give a nontrivial sum of positive group elements equal to zero. Strict separation of this compact convex set from zero gives a real linear functional positive on all these vectors; a small rational perturbation keeps the finitely many strict inequalities, and clearing denominators gives an integer-valued functional. If there are no strict comparisons,  $\ell = 0$  works.  $\square$

**Theorem 11.7** (Finite-support solutions specialize to arcs [05]). *Suppose  $\mathbf{x} \in \mathbf{Oz}^n$  has finite normal-form supports and satisfies a finite polynomial system over  $\mathbb{Z}$ . There are real polynomials  $f_1(T), \dots, f_n(T)$  satisfying the same polynomial identities with  $f_i(0) = \text{ct}(x_i)$ , and every nonconstant  $x_i$  gives a nonconstant  $f_i$ . A prescribed finite list of nonzero polynomial expressions in  $\mathbf{x}$  can be kept nonzero, with the same eventual signs as  $T \rightarrow +\infty$  as their surreal signs.*

*Proof.* Let  $E$  contain zero, all exponents of the coordinates, and all exponents of the prescribed expressions after expansion; it is finite. Let  $H$  be its integer span and choose  $\ell$  by lemma 11.6. The group-algebra map  $\mathbb{R}[H] \rightarrow \mathbb{R}[T, T^{-1}]$ ,  $\omega^h \mapsto T^{\ell(h)}$ , is a ring homomorphism. Positive coordinate exponents map to positive integers, so coordinate images are polynomials with the same constant coefficients; injectivity of  $\ell$  on  $E$  prevents cancellation of distinct supported terms, in particular of leading terms. Polynomial identities are preserved, and each prescribed expression keeps its leading coefficient and hence its eventual sign for large positive  $T$ .  $\square$

With proposition 11.1 (substituting  $t \in \Pi$  for  $T$ ), this gives an exact equivalence between the existence of a finite-support nonconstant solution and of a nonconstant real polynomial arc with integer constant coordinates. It does not turn infinite-support solutions into polynomial arcs.

For a finite-support implementation one may store an element as a map from exact exponents to nonzero exact real coefficients; the essential operations are coefficientwise addition, exponent addition in convolution, combination of equal exponents, and deciding integrality of the constant coefficient. Rational exponents with real algebraic coefficients give a practical decidable representation; general symbolic reals or arbitrary surreal exponents need further equality and order oracles. Useful exact certificates are polynomial identities, Bézout witnesses such as (11.2), standard-modulus quotients, and finite normal-form obstructions such as the first negative term in (12.2) [05].

## 12 Equations with omnific coefficients

For  $P \in \mathbf{Oz}[X_1, \dots, X_n]$  let  $\overline{P} \in \mathbb{Z}[X_1, \dots, X_n]$  apply  $\text{ct}$  to every coefficient. Then [02, 05]

$$\text{ct}(P(\mathbf{x})) = \overline{P}(\text{ct } x_1, \dots, \text{ct } x_n), \quad P(\mathbf{x}) = 0 \implies \overline{P}(\text{ct } \mathbf{x}) = 0.$$

The reverse implication need not hold, because the coefficients are no longer fixed by  $\text{ct}$ .

### 12.1 Quadratic equations

**Theorem 12.1** (Square-discriminant criterion [02]). *For  $b, c \in \mathbf{Oz}$ , the equation  $x^2 + bx + c = 0$  has a root in  $\mathbf{Oz}$  if and only if  $\Delta = b^2 - 4c$  is a square in  $\mathbf{Oz}$ . If  $y^2 = \Delta$  with  $y \in \mathbf{Oz}$ , its roots are  $(-b \pm y)/2$ , and both are omnific.*

*Proof.* A root gives  $(2x + b)^2 = \Delta$ . Conversely, suppose  $y^2 = \Delta$ . Specializing and reducing modulo 4 gives  $\text{ct}(y)^2 \equiv \text{ct}(b)^2 \pmod{4}$ . Integer squares are 0 or 1 modulo 4, so  $\text{ct}(y)$  and  $\text{ct}(b)$  have the same parity, and [theorem 3.1](#) shows that  $-b \pm y$  are divisible by two in  $\mathbf{Oz}$ . The usual identity verifies the roots.  $\square$

Testing whether a nonnegative surreal has an omnific square root is not the same as testing whether it has a surreal square root: real closedness gives the latter, while nonnegative support and an integral constant must still be checked for the former.

**Example 12.2** (Residual and leading tests that do not lift [01, 02, 05]). The equation  $x^2 = \omega^2 + 1$  has no solution in  $\mathbf{Oz}$  ([proposition 4.12](#)). Nevertheless its residual equation  $\text{ct}(x)^2 = 1$  is solvable, and the leading terms allow degree 1 and leading coefficient  $\pm 1$ . The positive root in  $\mathbf{No}$  has the formal binomial expansion

$$\sqrt{\omega^2 + 1} = \omega + \frac{1}{2}\omega^{-1} - \frac{1}{8}\omega^{-3} + \frac{1}{16}\omega^{-5} - \dots,$$

whose negative exponents exclude it from  $\mathbf{Oz}$ . This is a support-controlled Hahn expansion, not a claim of convergence in the surreal order topology. In contrast,  $\omega^2 + 2\omega + 1 = (\omega + 1)^2$  is an omnific square.

## 12.2 Initial forms

**Proposition 12.3** (Initial forms of an omnific solution [01, 02]). *Let  $P = \sum_{\alpha} a_{\alpha} \mathbf{X}^{\alpha} \in \mathbf{Oz}[X_1, \dots, X_n]$  be nonzero, let  $\mathbf{x} \in \mathbf{Oz}^n$  have all coordinates nonzero, and write  $\gamma_i = \deg_{\omega}(x_i)$ ,  $r_i = \text{lc}(x_i)$ . Discarding zero coefficients, put*

$$W = \max_{\alpha} \left( \deg_{\omega}(a_{\alpha}) + \sum_i \alpha_i \gamma_i \right).$$

*If  $P(\mathbf{x}) = 0$ , then*

$$\sum_{\deg_{\omega}(a_{\alpha}) + \sum_i \alpha_i \gamma_i = W} \text{lc}(a_{\alpha}) \prod_i r_i^{\alpha_i} = 0, \quad (12.1)$$

*and in particular the maximum is attained by at least two monomials.*

*Proof.* By [lemma 2.4](#), each evaluated monomial has degree  $\deg_{\omega}(a_{\alpha}) + \sum_i \alpha_i \gamma_i$  and leading coefficient  $\text{lc}(a_{\alpha}) \prod_i r_i^{\alpha_i}$ . The coefficient of  $P(\mathbf{x})$  at  $\omega^W$  is exactly the sum in (12.1), and it must vanish; a single maximizer would contribute a nonzero product.  $\square$

Variables assigned zero can first be substituted out. For ordinary coefficients this is a weighted initial polynomial over  $\mathbb{R}$  with surreal weights. The constant-coefficient equation and the initial equation constrain opposite ends of a normal form; they are complementary necessary tests, not a lifting criterion, as [example 12.2](#) shows: the missing information lies in the full tail of a potential root.

## 12.3 Two-term roots and failed lifting

**Proposition 12.4** (A two-term root obstruction [05]). *Let  $a \in \mathbb{R}_{>0}$ ,  $\gamma > 0$ ,  $b \in \mathbb{Z}$ , and let  $n \geq 2$  be ordinary. The equation  $X^n = a\omega^{\gamma} + b$  has a solution in  $\mathbf{Oz}$  if and only if  $b = 0$ . When  $b = 0$  its positive solution is  $a^{1/n}\omega^{\gamma/n}$ , together with its negative when  $n$  is even.*

*Proof.* The right side is positive and infinite. Its positive  $n$ th root in  $\mathbf{No}$  has the formal binomial expansion

$$a^{1/n} \omega^{\gamma/n} \sum_{k=0}^{\infty} \binom{1/n}{k} \left(\frac{b}{a}\right)^k \omega^{-k\gamma}, \quad (12.2)$$

whose exponents strictly decrease; the formal power-series binomial identity shows it is an  $n$ th root in the Hahn field, positivity gives uniqueness among positive roots, and in an ordered field the only other real root is its negative when  $n$  is even. If  $b \neq 0$ , the  $k = 1$  coefficient is nonzero at the exponent  $\gamma/n - \gamma < 0$ , which no other term has, so neither root is omnific. If  $b = 0$  the root is the single positive monomial displayed.  $\square$

**Example 12.5** (A simple residue root need not lift [05]). Let  $f(T) = T^2 - T - \omega$ . Modulo  $\Pi$  it becomes  $T(T - 1)$ , and the residue root 0 is simple, with derivative  $-1$ , a unit. Nevertheless  $f$  has no root in  $\mathbf{Oz}$ : its roots in  $\mathbf{No}$  are

$$\frac{1}{2} \pm (\omega^{1/2} + \frac{1}{8}\omega^{-1/2} - \frac{1}{128}\omega^{-3/2} + \dots),$$

with constant coefficient  $\frac{1}{2}$  and negative exponents. Hensel-style intuition fails for  $\mathbf{Oz} \rightarrow \mathbb{Z}$ : the kernel  $\Pi$  is not contained in the Jacobson radical, since  $-\omega \in \Pi$  but  $1 + \omega$  is not a unit, and  $\Pi^2 = \Pi$ , so there is no decreasing nilpotent-error filtration. This is a failure of the simple-root lifting property itself.

For general coefficients, a viable approach must inspect actual normal-form supports. Matching constant terms is necessary, and occasionally useful, but is not a sufficient solvability criterion ([question 14.6](#)).

## 13 Fixed Hahn workspaces and a formalization route

### 13.1 Which results are local in the exponent group?

For a set-sized ordered abelian group  $\Gamma$ , work in a Hahn field whose monomials are written  $\omega^\gamma$  with reverse-well-ordered supports, and define [01, 02]

$$\mathbf{Oz}_\Gamma = \mathbb{Z} + \left\{ \sum_{\gamma > 0} r_\gamma \omega^\gamma \right\}, \quad \mathcal{B}_\Gamma = \mathbb{R} + \left\{ \sum_{\gamma > 0} r_\gamma \omega^\gamma \right\}.$$

The displayed sums are elements of the ambient Hahn field with the specified support restrictions; this is a convention, not an identification of every abstract group element with a surreal exponent. When an ordered embedding of  $\Gamma$  into  $\mathbf{No}$  is fixed, it gives the corresponding surreal workspace.

**Local results.** The ring decompositions, the unit lemma, the retraction, finite quotients, equational transfer, the Smith criterion, constant-product and decomposable rigidity, binary, Pell and norm rigidity, the quadratic constructions, the quartic guards, and all explicit polynomial identities apply to  $\mathbf{Oz}_\Gamma$  [01, 02]. Their proofs use finite algebra, leading exponents and real coefficients, not the availability of a new exponent outside  $\Gamma$ . The derivation arguments of [section 7](#) take place in the workspace of the rational span of the relevant supports, so they apply to  $\mathbf{Oz}_\Gamma$  after passing to the divisible hull of  $\Gamma$  [05]. If  $\Gamma \neq 0$ , its positive cone provides nonconstant parameters for the infinite-family statements; the case  $\Gamma = 0$  is  $\mathbf{Oz}_\Gamma = \mathbb{Z}$ . The Lorentz family uses only  $t$ ,  $t^2$  and real scalars, so one positive monomial already produces examples. If the coefficient field is restricted below  $\mathbb{R}$ , the general indefinite lifting additionally needs the chosen real vectors  $e, u$  to be available, or an explicit scalar extension [02].

**Full-class results.** The common-divisor theorem for an arbitrary set of purely infinite elements, the monomial clearing theorem,  $\Pi^2 = \Pi$ , the non-generation of  $\Pi$  by a set,  $\text{Frac}(\mathbf{Oz}) = \mathbf{No}$ , the general pair without a gcd ([theorem 4.10](#)) and the set-sized quotient theorem ([cited theorem 3.6](#)) require bounds that may not exist in  $\Gamma$ . For instance, in  $\Gamma = \mathbb{R}$  the set  $\{1/k : k \geq 1\}$  has no strictly positive real lower bound, whereas in  $\mathbf{No}$  it has one. A support in  $\Gamma$  may also be unbounded below in  $\Gamma$ . These conclusions are not asserted for  $\mathbf{Oz}_\Gamma$ ; in particular one must not infer  $\Pi_\Gamma^2 = \Pi_\Gamma$  by copying the proof for  $\mathbf{No}$  [01, 02, 05]. Proofs that stay inside a workspace must be distinguished from proofs that enlarge it.

### 13.2 A formalization route

No part of this report has been formalized. At the placement commit the repository contains no Lean declaration about omnific integers. The three sources proposed overlapping routes; the following merges them. All module names are suggestions, not claims that such files or declarations exist.

| Proposed layer                                 | Main obligations                                                                                                                                                                                                                                            |
|------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>Omnific.Basic</code> [01, 02, 05]        | In a real Hahn field over a set-sized ordered divisible group: the nonnegative-support subring, the integer-constant subring, the retraction $ct$ from the zero-exponent convolution formula, and the finite-element criterion.                             |
| <code>Omnific.Residue</code> [01, 02]          | $n\mathbf{Oz} = \Pi + n\mathbb{Z}$ , finite quotients, preservation of $\Pi$ ; source 02 suggests proving these once for any ring retraction onto $\mathbb{Z}$ with an integer-divisible kernel.                                                            |
| <code>Omnific.Rigidity</code> [01, 02, 05]     | Additive leading degree, units of the real and complex support rings, the exact decomposable fibers (finite linear algebra plus coefficient extensionality), binary and norm rigidity.                                                                      |
| <code>Omnific.Derivation</code> [05]           | Euler derivations from a $\mathbb{Q}$ -linear functional (Leibniz from finite convolution fibers), the quotient step for separated powers, comaximality of powers, and Wronskian degree bounds.                                                             |
| <code>Omnific.Quadratic</code> [01, 02]        | The finite bilinear identities for $U_t$ over any field of characteristic zero, the isotropic-vector choices, and the level classification.                                                                                                                 |
| <code>Omnific.IntegralityGuard</code> [01, 05] | Pell rigidity with <code>Nat.sum_four_squares</code> (source 05's six-auxiliary form); source 01's five-auxiliary form needs a three-square theorem, which Mathlib lacks.                                                                                   |
| <code>Omnific.Global</code> [01, 02, 05]       | The small-cut theorem for support bounds, common monomial divisibility and global clearing, kept separate and with explicit smallness (universe) hypotheses; the upper bound for clearing and the positive lower bound for idempotence are distinct lemmas. |

Source 05 points to the normal-form bridge (`docs/NORMAL_FORM_BRIDGE.md`) and the modules `SmallNormalFormFieldEquiv.lean`, `SmallNormalFormHahnEmbedding.lean`, `SmallNormalFormWorkspace.lean` and `SignSequenceWorkspace.lean` under `Surrea`

1/Foundations/ as the transfer route to the actual surreal carrier; all four exist at the placement commit (with `SignSequenceWorkspaceExponents.lean`), and their signatures must be checked before use. The imported logical theorems (Tarski’s quantifier elimination, MRDP) remain separate dependencies; formalizing the retraction does not formalize them. The transvection identity is finite bilinear algebra with no dependence on birthday recursion, and the only hard number-theoretic input to the guards is the three- or four-square theorem.

## 14 Questions and synthesis

The sources list continuation problems that their proofs do not decide. None of them claims that these problems are open in the literature, or that the list is exhaustive. Two were settled inside the batch of manuscripts from which this report was built.

### 14.1 Questions settled by source 05

**Coprime exponents and  $Y^2 = X^3 + 1$  (source 01).** Source 01 asked which curves such as  $Y^2 = X^3 + 1$  have genuinely infinite omnific points, and proved rigidity of  $x^m - y^n = c$  only for  $\gcd(m, n) > 1$  ([remark 6.9](#)). [Theorem 7.4](#) of source 05 answers this for every separated equation  $ax^m + by^n = c$  with  $abc \neq 0$  and  $m, n \geq 2$ ; in particular  $Y^2 = X^3 + 1$  and  $x^2 - y^3 = 1$  have only ordinary omnific points ([corollary 7.6](#)). The general part of source 01’s question, a criterion for arbitrary affine curves, remains open as [question 14.1](#).

**Unimodular Fermat (sources 01 and 02).** Source 01 asked for the classification of primitive or unimodular omnific solutions of  $X^n + Y^n = Z^n$  with nonzero coordinates,  $n \geq 3$ ; source 02 asked which homogeneous systems admit nonzero omnific solutions generating **Oz**. [Theorem 7.9](#) and [corollary 7.10](#) of source 05 settle the unimodular Fermat case: there are none. The primitive case (no common nonunit divisor) remains open as [question 14.2](#), and source 02’s question for other homogeneous systems remains open as [question 14.3](#).

### 14.2 Open questions

**Question 14.1** (Rigid affine varieties and curves [01, 02, 05]). For which affine varieties  $X$  over  $\mathbb{Z}$  does  $X(\mathbf{Oz}) = X(\mathbb{Z})$  hold? For which affine curves does every omnific point have ordinary coordinates? A natural next test is  $Y^2 = X^3 + aX + b$  with  $a \neq 0$ , outside [theorem 7.4](#).

[Theorems 6.2](#), [6.3](#) and [7.4](#) answer this for nonzero decomposable levels with trivial common kernel and for separated equations, [theorems 9.2](#) and [9.6](#) for bounded loci and systems with no real point at infinity, and [theorem 8.1](#) and polynomial arcs give large opposing classes. A satisfactory criterion should separate polynomial parametrization, torus rigidity and genuinely algebraic branches [01]; it needs either a new degree inequality or a structural analysis of possible supports [05]. No claim is made that all other fibers are rigid, or that all nontrivial fibers come from real polynomial curves [02].

**Question 14.2** (Primitive Fermat triples [01, 05]). For  $n \geq 3$ , can the unit-ideal hypothesis of [theorem 7.9](#) be weakened to “no common nonunit divisor”? That is, are there primitive, non-unimodular omnific solutions of  $X^n + Y^n = Z^n$  with nonzero coordinates?

The answer cannot be read off from ordinary integer intuition; it needs an explicit relation between common divisors, ideal generation and the applicable factorization theory. [Theorem 7.9](#)

deliberately avoids any such relation, and the binary nonzero-fiber theorem does not directly answer a three-variable homogeneous question [01].

**Question 14.3** (Primitive homogeneous solutions [02]). Which homogeneous systems admit a nonzero omnific solution whose coordinates generate  $\mathbf{Oz}$ , and how does this depend on the chosen integral model of the projective variety?

The real-solvability criterion [theorem 11.2](#) omits unimodularity, and [theorem 11.5](#) exhibits the gap. A gcd normalization cannot be assumed, by [theorem 4.10](#). Fermat equations of exponent at least three have none ([corollary 7.10](#)); the exponent-two Pythagorean and indefinite quadratic cases have many ([proposition 11.4](#), (8.4)).

**Question 14.4** (An existential definition of  $\Pi$  [01]). Is there an existential definition of  $\Pi$  over  $\mathbf{Oz}$  with ordinary integer parameters? More generally, which fibers or subideals of  $\Pi$  are Diophantine? Retraction stability is necessary but far from sufficient.

**Question 14.5** (Complexity of the integrality guard [01]). Can the number of auxiliary variables be reduced below five, with a single quartic equation, or can a meaningful lower bound be proved? The present constructions give explicit upper bounds only (five in [theorem 10.3](#), six in [theorem 10.4](#)); other rigid arithmetic families might serve as finite bounds.

**Question 14.6** (Roots with omnific coefficients [01, 02, 05]). For polynomials with finitely represented omnific coefficients, which root questions in a chosen support class are decidable? Can one give tractable necessary and sufficient support criteria for a root of a polynomial in  $\mathbf{Oz}[T]$  to be omnific?

[Theorem 12.1](#) reduces the monic quadratic case to square membership, [proposition 12.4](#) settles two-term binomials, and [example 12.5](#) rules out a naive lifting theorem; multiterm cancellation needs separate analysis. Any effective claim must specify the coefficients, the exponent representation, and whether workspace enlargement is permitted. An unrestricted algorithm for ordinary-coefficient affine existence would contradict [corollary 5.2](#).

**Question 14.7** (The exact size boundary [05]). Which set-sized exponent groups or universe-relative constructions admit analogues of the common-monomial and set-sized quotient theorems?

The sibling report `docs/surreal/set-sized-quotients-of-omnific-integers` contains cardinal-threshold results on this question for specific set-sized integer parts.

**Question 14.8** (Effective finite-support discovery [05]). How far can a search for polynomial arcs and exact Bézout certificates, suggested by [theorem 11.7](#), go before transfinite support constructions are needed? It can find witnesses and expose obstructions, but it cannot become a complete decision procedure for integer-coefficient systems, by [corollary 5.2](#).

### 14.3 Synthesis

The arithmetic of  $\mathbf{Oz}$  is governed by three simultaneous facts:

$$\mathbf{Oz}/\Pi = \mathbb{Z}, \quad \mathbf{Oz}^\times = \{\pm 1\}, \quad \text{Frac}(\mathbf{Oz}) = \mathbf{No}.$$

The first makes ordinary congruences and equation-existence questions look classical. The second imposes powerful rigidity on nonzero product fibers. The third makes projective representation flexible while saying little about primitive representatives. Growth degree,

support and chosen Euler derivations then govern whether individual solutions escape the ordinary integers [05].

For Diophantine equations the key divide is not only between finite and infinite numbers. It is between *equation existence*, which the retraction controls; *multiplicative, differential or bounded rigidity*, which can force every coordinate to be ordinary; and *polynomial flexibility*, which creates infinite points without any infinitesimal tail [02]. The difference between a Pell conic and a three-variable indefinite quadric is structural: binary factorization forces constants, while an additional indefinite dimension supports finite polynomial isometries. The guards use the rigid side of this dichotomy to recover the ordinary integers inside  $\mathbf{Oz}$ . Omnific integers preserve much more of ordinary equational existence than their size suggests, and much less of ordinary quantified arithmetic than their discreteness suggests [05]. None of this requires unique factorization, a Euclidean termination principle, or Conway’s refinement property.

## 14.4 Relations to neighbouring reports

- `docs/foundations-and-computation/foundations`: defines  $\mathbf{Oz} = \Pi \oplus \mathbb{Z}$  (`found:eq:omnific`, subsection `found:sub:omnific`) with the same letter  $\Pi$ ; this report develops the arithmetic of that ring.
- `docs/surcomplex/trigonometry`: writes the purely infinite part with  $t$ -exponents of the opposite sign (`trigonometry:eq:split`; translation in [section 1.4](#)); its integer-part construction agrees with [theorem 2.7](#), and its identity  $2\pi\mathbf{Oz} = \Pi + 2\pi\mathbb{Z}$  (`trigonometry:eq:periodclass`) rests on the same fact  $r\Pi = \Pi$  for real  $r \neq 0$  as [theorem 3.1](#).
- `docs/surcomplex/gamma-and-zeta-functions`: the parity statements of `gz:cor:canonical-zeros` ( $-\omega/2 \in \mathbf{Oz}$ ,  $-\omega - 1 \notin 2\mathbf{Oz}$ ) are instances of [theorem 3.1](#); its statement that the internal integers of an ultrapower are not  $\mathbf{Oz}$  is consistent with [theorem 10.9](#) and [remark 10.10](#).
- `docs/surreal/transcendence-over-bounded-support`: its hypothesis `bst:hyp:gcd` that the bounded-support ring is a GCD domain concerns a different ring, whose units are the nonzero monomials; [theorem 4.10](#) concerns  $\mathbf{Oz}$ , whose units are  $\pm 1$ . Neither statement bears on the other.
- `docs/surreal/set-sized-quotients-of-omnific-integers`: contains a more general coefficient-ring version of the universal set-sized quotient theorem ([cited theorem 3.6](#)); its companion-source integration remains pending; its preliminaries overlap [sections 2 to 4](#).
- `docs/surreal/euclidean-three-space`: its theorem that class homomorphisms from  $\mathrm{SO}(3, \mathbf{No})$  to set-sized groups factor through the standard part (`e3:cut:thm:smallquotient`) is an analogous size phenomenon for groups, not the same theorem; by [corollary 9.4](#),  $\mathrm{SO}(3, \mathbf{Oz})$  is finite.

## A Provenance and merge record

### A.1 Sources

The report is built from three manuscripts of a batch of eleven placed in commit `be06fc8`; source numbers are the batch item numbers. All three are dated September 22, 2026 and pin repository commit `2cb9c0200af749fbbd27796c97d017e7f16fbf33`. None of the three was aware of the others. The manuscripts themselves are not shipped. Their code and

data are shipped under the prefixes `01-diophantine-geometry-`, `02-diophantine-` and `05-diophantine-rigidity-`; source 02's provenance note and source 05's build note are shipped as `02-diophantine-PROVENANCE.md` and `05-diophantine-rigidity-BUILD.md`. Source 01 delivered a checksum manifest, which is not shipped because it describes the delivered file names.

## A.2 Section map

| Section                                         | Sources                                                                                                                                                            |
|-------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <a href="#">1</a> Scope and conventions         | 01 base; 02 and 05 results; new notation and provenance text                                                                                                       |
| <a href="#">2</a> Rings and integer part        | 01 base; 05 degree of sums, workspaces, univariate over $\mathbb{R}$ ; 02 examples                                                                                 |
| <a href="#">3</a> Residues and quotients        | 01 base; 02 mixed gcds and completions; 05 $p$ -adic orders, divisors of integers, CRT; 05's set-sized quotient theorem cited                                      |
| <a href="#">4</a> Full-class support            | 01 base; 02 nilpotent tests, common multiples, general no-gcd pair; 05 witness $\sqrt{2}$ and prime examples                                                       |
| <a href="#">5</a> Transfer and linear equations | 01 base; 02 fibers and examples; 05 rational-basis form                                                                                                            |
| <a href="#">6</a> Constant products             | 02 exact fibers; 05 binary form over $\mathbb{C}$ , $D = 2$ classification, conics; 01 zero Pell fiber, $\gcd > 1$ powers, norm forms with any $\mathbb{Q}$ -basis |
| <a href="#">7</a> Derivations                   | 05 entirely; merge remark <a href="#">7.5</a>                                                                                                                      |
| <a href="#">8</a> Quadratic dichotomy           | 01 base; 02 skewness, determinant, Lorentz matrix, unipotent subgroup                                                                                              |
| <a href="#">9</a> Bounded geometry              | 02 semialgebraic theorem; 01 positive bounds, symmetric matrices, no real point at infinity; 05 even powers                                                        |
| <a href="#">10</a> Defining $\mathbb{Z}$        | 01 three-square guard; 05 four-square definition and induction; merge remark <a href="#">10.10</a>                                                                 |
| <a href="#">11</a> Families and primitivity     | 01 base; 02 sign conditions and decidability; 05 witness and finite-support specialization                                                                         |
| <a href="#">12</a> Omnific coefficients         | 02 quadratic criterion and initial forms (also 01); 05 binomials and failed lifting                                                                                |
| <a href="#">13, 14</a>                          | All three; settled questions recorded in <a href="#">section 14.1</a>                                                                                              |

## A.3 Where the merge chose

1. *Base text.* Source 01, which has the weakest hypotheses on most shared statements (Pell with  $D \neq 0$ , norm forms for any  $\mathbb{Q}$ -basis and  $c \in \mathbb{Q}^\times$ , the only exhaustive quadratic dichotomy, the guard uniform in  $n$ , the QE-free homogeneous criterion, the localization discussion).
2. *Decomposable fibers.* Source 02's exact-fiber theorem is printed, with source 01's rank- $n$  theorem as case (b) of [theorem 6.2](#).

3. *Binary forms*. Source 05's complex form is printed; sources 01 and 02 proved the integer case.
4. *Power equations*. Source 05's separated-power theorem replaces source 01's  $\gcd > 1$  corollary, which is kept as [remark 6.9](#).
5. *Isotropic vectors*. Source 02's hypothesis  $a \neq 0$  replaces source 01's  $q(a) \neq 0$  in [lemma 8.3](#).
6. *Pairs without gcd*. Source 02's general theorem is printed; source 01's degree proof is kept as a second route ([remark 4.11](#)) because it localizes to divisible workspaces.
7. *Homogeneous systems*. Source 01's elementary proof and source 02's Tarski proof are both kept, since only the latter handles sign conditions and decidability.
8. *Uniqueness of ct*. Source 01's divisibility proof is printed; source 05 derives it from the set-sized quotient theorem, which is quoted and whose proof belongs to the sibling report.
9. *Guards*. Both are printed: source 01's (five auxiliaries, three squares) and source 05's (six auxiliaries, four squares).
10. *Additions*. Two remarks carry the tag [merge]: [remark 7.5](#) (a direct identity giving the divisibility in the separated-power proof) and [remark 10.10](#) (existential induction fails in the pure ring language). Both have complete proofs.

**Printed once.** The floor formula, retraction, congruences, finite quotients and completions, transfer, Hilbert's tenth problem, the Smith criterion, units, the Euclid example, non-atomicity, the ACC chain,  $\Pi^2 = \Pi$  and non-generation,  $\text{Frac}(\mathbf{Oz}) = \mathbf{No}$ , the Pythagorean family, the Fermat scaling,  $x^2 = \omega^2 + 1$ , the orthogonal group, the nilpotent isometry, the torus remark, and the failure of four squares for  $\omega - 1$  appear in two or three sources and are printed once, with all sources named.

## A.4 Renamed symbols

Table 1: Symbols renamed from each source. No normalization changed except where stated.

| Source | Source symbol                                 | This report                                    |
|--------|-----------------------------------------------|------------------------------------------------|
| 01     | $A$ (omnific integers)                        | $\mathbf{Oz}$                                  |
| 01     | $\mathcal{I}$ (purely infinite ideal)         | $\Pi$                                          |
| 01     | $\text{Int}, \text{Int}(x)$                   | $\text{Std}_n, \text{Std}$                     |
| 01     | $G_n$ (guard polynomial)                      | $\Psi_n$ ( $G$ is the Lorentz Gram matrix)     |
| 01     | $A_\Gamma, B_\Gamma$                          | $\mathbf{Oz}_\Gamma, \mathcal{B}_\Gamma$       |
| 01     | $P$ (purely infinite part in the floor)       | $x_{>0}$                                       |
| 01     | $\epsilon$ (common-divisor exponent)          | $\delta$                                       |
| 01     | $\alpha = \sqrt{2} - 1$                       | $\rho$                                         |
| 01     | $\delta = \deg_\omega d$ (gcd proof)          | $\theta$                                       |
| 01     | $E$ (hyperboloid coefficient)                 | $D'$                                           |
| 01     | $T_t$ (transvection)                          | $U_t$                                          |
| 01     | $\lambda_\alpha, c_\alpha, M$ (initial forms) | $\deg_\omega a_\alpha, \text{lc } a_\alpha, W$ |
| 01     | $\lambda$ (scalar in decomposable forms)      | absorbed into one factor                       |

| Source | Source symbol                                                                                  | This report                                                                        |
|--------|------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------|
| 01     | $f$ (endomorphism)                                                                             | $\psi$                                                                             |
| 02     | $\mathcal{I}, \mathcal{I}_{\mathbb{C}}$                                                        | $\Pi, \Pi_{\mathbb{C}}$                                                            |
| 02     | $\mathcal{A}_{\mathbb{R}}, \mathcal{A}_{\mathbb{C}}$                                           | $\mathcal{B}_{\mathbb{R}}, \mathcal{B}_{\mathbb{C}}$                               |
| 02     | $d(f)$                                                                                         | $\deg_{\omega} f$                                                                  |
| 02     | $Q, B$ (quadratic and bilinear form)                                                           | $q, B$                                                                             |
| 02     | $a, q = Q(a)$ (lemma vectors)                                                                  | $u, k = q(u)$                                                                      |
| 02     | $u$ (integer point)                                                                            | $\mathbf{a}$                                                                       |
| 02     | $U_T, T, S$                                                                                    | $U_t, t, s$                                                                        |
| 02     | $J = \text{diag}(1, -1, -1)$                                                                   | $G$                                                                                |
| 02     | $J$ (index set)                                                                                | sets $S$ of elements                                                               |
| 02     | $J$ (ideal)                                                                                    | $\mathbf{a}$ or $\mathbf{n}$                                                       |
| 02     | $A$ (integer matrix)                                                                           | $M$                                                                                |
| 02     | $K$ (common real kernel)                                                                       | $\mathcal{K}$                                                                      |
| 02     | $\eta$ (infinitesimal part), $P$ (purely infinite part)                                        | $\varepsilon, x_{>0}$                                                              |
| 02     | $s = \sqrt{2} - 1$                                                                             | $\rho$                                                                             |
| 02     | $T$ (in the no-gcd theorem)                                                                    | $t$                                                                                |
| 05     | $\mathcal{I}$                                                                                  | $\Pi$                                                                              |
| 05     | $\mathcal{B}, \mathcal{B}_H, \mathcal{B}_{\mathbb{C},H}$                                       | $\mathcal{B}_{\mathbb{R}}, \mathcal{B}_{\mathbb{R},H}, \mathcal{B}_{\mathbb{C},H}$ |
| 05     | $A$ (generic ring; set-sized target)                                                           | $R; S$                                                                             |
| 05     | $D_{\lambda}, W_D$                                                                             | $\partial_{\lambda}, W_{\partial}$                                                 |
| 05     | $D, N$ (Pell)                                                                                  | $D, c$                                                                             |
| 05     | $p$ (Fermat exponent)                                                                          | $n$ ( $p$ is reserved for primes)                                                  |
| 05     | $a, b$ (ratio constants in the Fermat proof)                                                   | $\mu, \nu$                                                                         |
| 05     | $\deg_{\omega}, \lfloor \cdot \rfloor_{\mathbf{OZ}}, X_k, Y_k, \rho, \text{Std}, \Phi, x_{>0}$ | unchanged                                                                          |

## A.5 Corrections to the sources

1. Source 02 states Pell rigidity for  $D > 0$  (in its abstract, its introduction, its catalogue table and its Pell corollary); its proof uses only  $D \neq 0$ , as in sources 01 and 05. [Corollary 6.5](#) states  $D \neq 0$ .
2. Source 01’s abstract says that nonzero fibers of “nondegenerate binary forms” are rigid. Its theorem requires, and proves, the weaker hypothesis of at least two distinct projective linear factors, which in degree at least three is weaker than squarefreeness ( $X^2Y$  qualifies). The abstract of this report states the theorem’s hypothesis.
3. Source 02 cites Matiyasevich’s translation as *Soviet Math. Dokl.* 11 (1970), 354–358, and source 05 as 354–357. The range 354–357 is the one given, with MathSciNet number MR 258744, in the bibliography of Bhatt and Poonen’s notes *Diophantine sets*, and in the title of the review of the paper in the *Journal of Symbolic Logic*; it is used here.
4. Source 05’s proof of the set-sized quotient theorem invokes global choice to pick a cardinal; the Hartogs ordinal suffices ([remark 3.7](#)).
5. Source 01’s isotropic-vector lemma assumes  $q(a) \neq 0$  where  $a \neq 0$  suffices ([lemma 8.3](#)).

6. Source 05’s verification script checks the separated-power identity only for  $a = 1$ ,  $b = -1$ ; the general identity of [remark 7.5](#) was checked separately (symbolically, for  $2 \leq m, n \leq 7$ ) when this report was assembled. The shipped script is unchanged.
7. Source 01’s questions on  $Y^2 = X^3 + 1$  and on unimodular Fermat triples, and the Fermat case of source 02’s question on unimodular homogeneous solutions, are recorded as settled by source 05 ([section 14.1](#)); the remaining parts are re-scoped as open questions.
8. Source 02’s attribution of the nonzero omnific Fermat failure to MathOverflow comments could not be verified; it is printed as source 02’s attribution, marked unverified ([example 11.3](#)).

No mathematical error was found in any main proof of the three sources.

## A.6 Stale repository statements

All three sources inspected the repository at `2cb9c0200af749fbbd27796c97d017e7f16fbf33` through the GitHub connector, without a checkout or build.

1. Source 01 says that the repository’s trigonometry material already uses the omnific integer part. At the pin this was true but incomplete: the foundations report already defined  $\mathbf{Oz} = \Pi \oplus \mathbb{Z}$  (`found:eq:omnific`). The collection now also contains this report and the sibling report on set-sized quotients.
2. All three sources correctly found no report devoted to  $\mathbf{Oz}$  and no Lean declaration about it at the pin. The first half is no longer true; the second still is.
3. Sources 02 and 05 describe `docs/NORMAL_FORM_BRIDGE.md` and the workspace modules; these descriptions are still accurate at the placement commit.
4. The status of Conway’s refinement conjecture is recorded as of September 22, 2026 ([appendix D](#)); the transcendence-over-bounded-support report takes the same stance.

## B Non-claims by source

Every limitation, disclaimer and priority caveat stated by a source is kept. Most also appear at their point of use in the text; this appendix is the per-source record.

**Common to all three sources.** No independent peer review; no Lean or other proof-assistant verification; the finite scripts check finite identities and examples, not the theorems; the repository was not checked out or built; the Conway refinement announcement of September 18, 2026 was not audited, is not used, and is neither assumed nor claimed settled; no result assumes refinement, GCD or UFD properties; no priority claim is made.

### Source 01 (24 items).

1. Proving a statement is not a claim that nobody proved it before; no exhaustive historical-priority claim; the quadratic classification and the guard are named as its most distinctive constructions, found in the inspected sources in no such form.

2. Familiar mechanisms (coefficient retractions, unit arguments, polynomial parametrizations, nilpotent quadratic isometries) are not claimed as new; a focused search does not establish priority.
3. The script is a supplement, not a proof; it does not implement the surreal numbers.
4. Transfer does not include inequations or order conditions.
5. Transfer concerns existence, not cardinalities or complete solution sets.
6. The Hilbert's-tenth-problem statement concerns ordinary integer coefficients; omnific coefficients are a different question.
7. Thue levels: absence of additional omnific solutions only; no new proof of finiteness and no algorithm for ordinary solutions.
8. The  $\gcd > 1$  power argument does not classify coprime cases such as  $x^2 - y^3 = 1$  (now covered by source 05).
9. The torus remark does not say that every affine variety is constant.
10. The quadratic classification concerns represented nonzero levels; it does not decide representability or classify all infinite points.
11. The guard's five auxiliary variables are not claimed optimal.
12. The first-order definition of  $\Pi$  is not an existential definition.
13. Polynomial lifting does not license evaluating infinite power series.
14. Fermat scaling solutions are not primitive and resolve no primitive Fermat problem.
15. The rational-direction theorem concerns real projective points only.
16. Initial equations are necessary, not sufficient.
17. The converse of the no-real-point-at-infinity theorem is false.
18. Common divisors, clearing,  $\text{Frac}(\mathbf{Oz}) = \mathbf{No}$  and non-generation of  $\Pi$  are full-class and not asserted for fixed workspaces; clearing is representability, not primitivity.
19. The three-square theorem, MRDP and the normal-form foundations are imported, not reproved.
20. L'Innocente–Mantova's Theorem B is not used in any proof.
21. None of its arguments refutes the refinement property.
22. Proposed module names are proposals; no Lean files; not checked against a built repository.
23. A missing repository search match is not proof of absence.
24. Its questions are not all claimed open in the literature.

### Source 02 (22 items).

1. The definition, integer-part property and pathological factorization belong to established theory; the retraction and many consequences should not be advertised as new.
2. The literature search does not establish priority for its principal formulations (exact decomposable fibers, uniform quadratic lifting); independent rediscovery is likely; absence of an identical formulation is no evidence of priority.
3. AI-assisted; no peer review; no complete Lean verification; no Lean build.

4. The checks do not prove transfinite-support results, the Diophantine claims, or the imported quantifier-elimination and undecidability theorems.
5. Transfer says nothing about how many points lie above an integer point.
6. The common-multiple theorem is an existence theorem, not an algorithm; no nonzero element is divisible by every nonzero omnific integer.
7.  $\Pi/\Pi^2 = 0$  does not mean the infinite part is small.
8. The L’Innocente–Mantova prime is not reproved.
9. No effective enumeration of surreals is assumed; arbitrary class coefficients have no implicit finite encoding.
10. Nonvanishing and order do not transfer; the retraction is not an elementary embedding; residual solvability need not lift for non-integer coefficients.
11. The failure of four squares does not contradict open induction.
12. Residual and initial tests are not a complete lifting criterion.
13. The homogeneous theorem does not decide affine omnific equations.
14. The Fermat example is not a refutation of Fermat’s Last Theorem and not a novelty claim.
15. Clearing and homogeneous solvability do not answer the primitive-representative question.
16. Polynomial maps are not assumed to exhaust all constructions.
17. Binary and norm rigidity give no effective bounds for ordinary points.
18. Not every omnific linear group is finite.
19. The quadratic classification concerns the appearance of infinite points, conditional on ordinary solvability; not every point lies on the orbit; degenerate forms and the zero level are excluded from its statement.
20. Do not infer  $\Pi_G^2 = \Pi_G$  for fixed workspaces; restricted coefficient fields need the vectors  $e, u$  or a scalar extension.
21. Its questions: no claim that other fibers are rigid or come from curves; no general root-membership algorithm (one would contradict H10).
22. Formalization stages are proposals; formalizing the retraction does not formalize Tarski or MRDP; the repository review was targeted and the normal form theorems are used as mathematical inputs, not as an audit of the implementation.

**Source 05 (26 items).**

1. No general solution of omnific Diophantine equations and no effective presentation of every surreal.
2. No new proof of the theorems of Lagrange, Matiyasevich, Shepherdson or Wiles.
3. No historical-priority claim for the derived results.
4. Not peer reviewed; not formally verified; finite checks are not substitutes for proofs and do not check arbitrary supports or the class-theoretic pigeonhole argument.
5. The normal-form bridge implementation is not independently certified; the Lean implementation was not built or audited.
6. The substantial factorization theory of L’Innocente–Mantova is a separate achievement, not replaced by elementary arguments.

7. Transfer does not make fibers singletons.
8. The computational statement does not cover omnific coefficients.
9. Projective representability does not give unimodular representatives.
10. The Euler derivations are local and not canonical, and are not the Berarducci–Mantova derivation.
11. Full-class results are not asserted for a preassigned workspace.
12. The set-sized quotient theorem does not apply to class targets such as  $\mathbf{Oz}$ , and it does not say that endomorphisms are the identity.
13. The size conclusion for  $\mathbf{Oz}/(f)$  concerns the class ring, not a set-sized quotient.
14. There is no Pell solution at index  $\omega$ .
15. No Thue finiteness proof is contained in the reduction.
16. The conic statement covers central conics, not every quadratic equation.
17. Separated-power rigidity gives no bound or complete list of integer solutions; the script certifies no completeness.
18.  $y^2 = x^3 + ax + b$  with  $a \neq 0$  is outside the method, a genuine boundary.
19. Only fixed ordinary exponents are treated, not variable omnific exponents.
20. The Fermat theorem says nothing about triples that merely lack a common nonunit divisor, and assumes no gcd equivalence.
21. The first-order definition of  $\text{ct}$  is not existential.
22. The four-square theorem is used only on ordinary nonnegative integers.
23. Transcendence of infinite omnific integers does not imply algebraic independence of tuples.
24. Finite-support specialization does not turn infinite-support solutions into arcs; finite-support search is not a complete decision procedure; matching constant terms is not a sufficient solvability criterion.
25. Its questions are not an exhaustive list of established open problems.
26. Its formalization route is a proposal; declarations were not invented; the size-sensitive layer must not be generalized to fixed Hahn groups.

## C Dependencies and verification

### C.1 Imported inputs

The foundation imported from surreal theory is the existence, uniqueness, order and multiplication of set-supported Conway normal forms, the set-cut property, and the real closedness of  $\mathbf{No}$  [1, 2]. The other imported theorems are: MRDP (corollary 5.2); Tarski’s quantifier elimination (theorem 9.2, theorem 11.2(b)); the three-square theorem (lemma 10.2); Lagrange’s four-square theorem (theorem 10.4, corollary 5.2); Shepherdson’s theorem (open induction); the Fermat theorem of Wiles and Taylor–Wiles (corollary 7.10 only, after the rigidity theorem has reduced to ordinary integers); the algebraic closedness of  $\mathbb{C}$  (to factor binary forms; the algebraic closedness of a class-sized surcomplex field is not used); and, as context only, the primes of [3] and [4, Lemma 11.1]. All other deductions are proved here from these and finite algebra.

The principal dependency chains are:

normal forms  $\implies$  ct is a retraction  $\implies$  transfer and finite quotients;  
 normal forms  $\implies$  constant units in  $\mathcal{B}_{\mathbb{C}}$   $\implies$  decomposable, binary and norm rigidity;  
 local Euler derivations + binary rigidity  $\implies$  separated powers and unimodular Fermat;  
 Pell rigidity + three- or four-square theorem  $\implies$  quartic definitions of  $\mathbb{Z}$ ;  
 real quadratic linear algebra + support-ring algebra  $\implies$  quadratic-level dichotomy;  
 normal forms + set-cut bounds  $\implies$  common divisors and clearing.

The source-05 proofs that most deserve independent review are, by its own account, the separated-power theorem (divisibility in a quotient ring without unique factorization), the unimodular Fermat theorem (strict leading-term cancellation in the equal-degree Wronskian), and the set-sized quotient theorem (positivity of every exponent in the geometric quotient; printed in the sibling report).

## C.2 Finite checks

Three scripts are shipped in `code/`, each with its recorded output in `data/`; they need Python 3.9 or later and SymPy (tested with SymPy 1.14.0). They check finite identities and examples only.

- `01-diophantine-geometry-verification.py`: scalar identities (Pythagorean and hyperboloid identities and their Bézout certificates, the cubic norm by resultant, the Euclid ratio), 20 quadratic-isometry cases in dimensions 3 to 7, Pell indices 0 to 30, and 256 guard witnesses. It writes a JSON report, by default `verification_report.json` in the working directory, so it should be run with `-output` pointing elsewhere or on a copy; the shipped record is `01-diophantine-geometry-verification_report.json`.
- `02-diophantine-verify_examples.py`: 13 exact identity checks (Pythagorean, cubic norm two ways,  $N^3 = 0$ , skewness, Lorentz preservation, determinant, group law and inverse, the hyperboloid family, binomial coefficients, the Euclid recurrence) and 49 finite-support products; output only to the terminal, recorded in `02-diophantine-verification.txt`.
- `05-diophantine-rigidity-verify_examples.py`: nine groups (Pell recurrence, Pythagorean witness, the  $\rho$  identity, 64 separated-power identities, Fermat Wronskian identities for  $2 \leq p \leq 10$ , 16 finite telescoping identities, binomial root coefficients, quartic-definition certificates for  $|t| \leq 40$  by default, and five points on  $y^2 = x^3 + 1$ ); output to the terminal, recorded in `05-diophantine-rigidity-verification.txt`.

When this report was assembled all three passed with Python 3.14.4 and SymPy 1.14.0, and source 05's script also passed with `-standard-radius 200`. The outputs of sources 01 and 02 differ from the shipped records only in the recorded Python version (3.13.5 there); the output of source 05 is identical to its record.

## D Repository and literature audit

### D.1 Repository snapshots

All three sources pin `2cb9c0200af749fbbd27796c97d017e7f16fbf33` ("Merge origin/main after the inverse-sine series formalization"). Source 01 read the tree, the root README, the

documentation index, the trigonometry listing and its period-arithmetic source audit; source 02 read the README, the documentation index and the opening of the normal-form bridge note; source 05 read the documentation index and the bridge note. Each source says that this was a focused inspection through the GitHub connector, not a checkout, build, or review of every manuscript and declaration. The checks of their repository statements against the placement commit are in [appendix A](#).

## D.2 Literature checks

- L’Innocente–Mantova, arXiv:1710.07304v5: Theorem B states that  $\omega^{\sqrt{2}} + \omega + 1$  is prime in **Oz**, answering a question of Gonshor; their Section 1.3 reviews the earlier results on  $1 + \sum_k \omega^{1/k}$ ; their introduction records, citing Shepherdson, that the nonnegative part of **Oz** is a model of open induction. The citations of all three sources agree with this.
- Shiu (*Math. Gazette* 104 (2020)) and the Archive of Formal Proofs entry of Danilkin and Chevalier (2023) for the three-square theorem, and Ehrlich–Kaplan Lemma 11.1 (as also cited in the trigonometry report), were confirmed during placement.
- Jeřábek’s MLQ article on rigid models of Presburger arithmetic (source 01) is cited for background; it was not re-read for this report.
- Source 02’s MathOverflow attribution could not be checked ([example 11.3](#)).
- Matiyasevich’s page range is corrected as described in [appendix A](#).

## D.3 A dated factorization announcement

On September 18, 2026 Dan Abramov announced a Lean proof of Conway’s refinement conjecture [19, 20], noting explicitly the absence of independent mathematical verification. None of the sources audited that code or used the announced result. Accordingly this report does not repeat older descriptions of the refinement conjecture as a verified *currently open* problem, does not claim to settle it, and none of its proofs depends on its status.

## References

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